

PolyGraphs: Simulating Testimonial Injustice

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Artificial Intelligence Dissertation Project

AE2 - Dissertation

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Abstract

The PolyGraphs framework enables the modeling of epistemic networks, and the computing of simulations on these networks. This project utilizes the framework to investigate the effects of testimonial injustice on group knowledge, situating it within computational social epistemology. The simulation results suggest that nodes that enact testimonial injustice are epistemically disadvantaged, as compared to those that do not. However, there are cases where the overall network benefits from testimonial injustice due to its impact on network density, and the Zollman effect. Future work should focus on expanding the parameter space to increase the robustness of the results, and on developing a new custom operation to control for network density.

1. Introduction

1.1. Setup

Understanding the network model and simulations setup is essential to understanding this project. Like Zollman,¹ I model a network of scientists, where each scientist attempts to determine whether drug A or drug B is better. Drug A is old, and so its efficacy rate is known— drug A works 50% of the time, and the scientists know this. Drug B, however, is new— it works 50.1% of the time, but the scientists do not know this. Only scientists who believe that drug B is better give drug B to their patients, thus gathering new evidence for the efficacy rate of drug B, and sharing that evidence with their neighbors.

¹ K. Zollman, *The communication structure of epistemic communities*.

So, the nodes in the modeled networks are scientists, and the edges in the network describe where scientists are connected to one another by evidence-sharing. Each simulation step consists of evidence-gathering, evidence-sharing, and belief-updating. So, for one step, the scientists that believe drug B is better gather new evidence (successful trials : total trials), and share this evidence with the scientists that they are connected to. Then, at the end of the simulation step, each scientist in the network updates her individual belief, taking her current belief, the evidence she received at this step, and her certainty in that evidence. The simulation keeps taking steps like this until: 'A,' none of the scientists believe that drug B is better than drug A, 'B,' all of the scientists believe that drug B is better than drug A, or '?,' the max limit on simulation steps is reached (cut-short).

Testimonial injustice is modeled by splitting the scientists into groups. One group always contains scientists that enact testimonial injustice, and another always contains scientists that are the victims of testimonial injustice. When scientists in that first group receive evidence from scientists in that second group, they discount said evidence by updating on it with decreased certainty. So, some scientists in the network discount testimony due to the group-membership of the testifier, thus modeling testimonial injustice.

1.2. Scope

The parameter space and network structure of my model is rather narrow, meaning that results are not very robust. I only run simulations on complete networks (where every node is connected to every other node), and at sizes, 16 and 64. I check for when there are 2 groups, and for when there is a third, neutral group (scientists that neither discount evidence nor have their evidence discounted). Finally, I check two certainty values for when testimonial injustice occurs, 0.5 and 0.75. Certainty is always set to 1 outside of instances of testimonial injustice (no evidence-discounting). I check all combinations of these parameters, totaling 8 simulation configurations. For all configurations, epsilon is set to 0.001, meaning that drug B is always only marginally more effective than drug A.

The simulated group-dynamics are similarly narrow, only representing a fraction of all those dynamics where testimonial injustice might occur. For example, the groups are mutually exclusive—a victim of testimonial injustice never enacts testimonial injustice. Further, the group sizes are always equal, and the group labels, always static.

Though I did not arrive at an explicit hypothesis before analyzing the simulation results, I did expect to find that testimonial injustice would be epistemically harmful. Therefore, much of this project's data analysis is exploratory; hypotheses are formulated and tested along the way.

1.3. Terms

Testimonial-injustice: A subcategory of epistemic injustice; e.g., Inquirer X enacts testimonial injustice in allowing prejudice against testifier Y to reduce the level of credibility she assigns to testifier Y, or to testifier Y's testimony²

Size: The number of nodes in a network

Groups: The number of groups in a network

Action: The outcome of the simulation, 'A,' 'B,' or '?,' as described in section 1.1

² M. Fricker, "Epistemic Injustice," *Epistemic Injustice: Power and the Ethics of Knowing*.

Steps: The number of steps a simulation took until it reached an outcome

Epsilon: The difference in efficacy rates of drug A and drug B

Certainty: The level of certainty that a node has regarding the evidence it is updating on

CIP: Shorthand for certainty-in-case-of-prejudice; the certainty value used for instances of testimonial injustice; sometimes seen as ‘mistrust’

Prejudiced: The group label for nodes that enact testimonial injustice

Prejudiced-against: The group label for nodes that are victims of testimonial injustice

Simulation-configuration: The combination of parameters used to run a simulation; sometimes written as, say, ‘16_2_75,’ where the first position indicates size, the second, groups, and the third, CIP

2. Literature Review

Computational Philosophy: Reflections on the PolyGraphs Project

The authors³ describe their project in computational social epistemology, PolyGraphs, roughly formulating their research question as “How ought we rationally to form opinions (i.e., beliefs), both individually and collectively? The authors attempt to answer this question by modeling a social network, and simulating the updating of beliefs for each member (node) of the network as that member attempts to discern the truth about some objective claim. They analyze the resulting synthetic data to answer the aforementioned research question. It is important to note, however, that the research question is normative, as it deals with how things ‘ought’ to be, and can therefore not be answered purely empirically, as might be possible in similar endeavors, such as those belonging to the computational social sciences.

The Misinformation Age: How False Beliefs Spread, Chapter 2 – Polarization and Conformity

The authors⁴ describe how evidence-sharing across a scientific community causes ambivalent effects on that community’s ability to converge on a true belief. Sharing evidence can be beneficial because it allows for a new belief to take hold in a way that would not be possible without shared evidence. However, the authors note that sharing evidence can also have ill-effects on belief within the community. Sometimes, the sharing of misleading evidence—evidence that suggests a false belief—can lead members of a community to abandon their true belief in favor of a new, false one. Therefore, the authors frame said ambivalent effects as a trade-off. They conclude that a member of some community might improve her beliefs in failing to communicate/ take shared evidence into consideration— the Zollman effect, named after Zollman’s initial modeling of the phenomenon.

Furthermore, The authors modify Zollman’s model by replacing Bayes’ rule with Jeffrey’s rule to account for human doubt in interpreting evidence. They calculate uncertainty based on the distance between beliefs of nodes (evidence-sending-node and evidence-receiving-node), resulting in a modeling of polarization, where distinct groups converge to different beliefs, eventually disregarding

³ B. Ball, A. Koliouisis, A. Mohanan, and M. Peacey, *Computational philosophy: reflections on the PolyGraphs project*.

⁴ C. O’Connor and James Owen Weatherall, *The Misinformation Age: How False Beliefs Spread*.

all evidence from the other group.

Epistemic Injustice: Power and the Ethics of Knowing

The author⁵ proposes a category of injustice that is epistemic in nature, aptly named ‘epistemic injustice’. She notes that many injustices might be imagined in which epistemology is relevant, but that some such instances might fail a necessary condition of epistemic injustice— that said injustice fundamentally wrongs someone “in their capacity as a knower.”

The author proposes two sub-categories of epistemic injustice— testimonial injustice and hermeneutical injustice. Inquirer X enacts testimonial injustice in allowing prejudice against testifier Y to reduce the level of credibility she assigns to testifier Y, or to testifier Y’s testimony. Hermeneutical injustice is caused by structural prejudice, and occurs when members of a group collectively lack the means of communication to allow for some testifier to express their knowledge. In instances of hermeneutical injustice, some testifier is wronged in their capacity as a knower not due to the particular prejudices of some inquirer, but due to “prejudicial flaws in shared resources for social interpretation.”

Network Science

The author⁶ provides background context on networks, describing them as groups of nodes connected by links. These links are directional in a directed network, such as in a network of airplane flights, where airports are nodes and flights are links, and they are unidirectional in an undirected network, such as the network of a social group, where nodes are friends and the link are friendships. The author also defines relevant vocabulary, defining a node’s degree as how many connections that node has, and a network’s sparsity as the measure of how much fewer links exist in that network than the network could have, if it were fully-connected/complete (every node has a link to every other node). The author points out that real networks (actual) tend to be sparse.

3. Work Undertaken

3.1. Framework Modification

In order to implement a group-based distrust mechanism on the PolyGraphs framework, I modified ‘hyperparameters.py’ and ‘graphs.py’ to support the grouping of nodes in a network. I added ‘groups’ (number of groups) to the class, ‘NetworkHyperParameters,’ so that the number of groups in the network can be specified alongside the other parameters taken by the PolyGraphs function, ‘pg.simulate()’. This would later enable me to generate simulations with a neutral third group, beyond the minimum two-group setup required for simulating testimonial injustice.

To actually group the nodes, I merely assign each node a label. If the aforementioned ‘groups’ parameter is set to 2, I equally distribute ‘0’ and ‘1’ labels, one such label for each node in the network. If the parameter is set to 3, I equally distribute ‘0,’ ‘1,’ and ‘2,’ one such label for each node in the network, and so on. If the network size (number of nodes in the network) is not divisible by the

⁵ M. Fricker, “Epistemic Injustice,” *Epistemic Injustice: Power and the Ethics of Knowing*.

⁶ A. Barabási, *Network science*.

number of groups, the remaining node is labeled '0', and if there are two remaining nodes, the second is labeled '1'. I figured that having an extra node in a group or two at some simulation configurations would not impact the results to a statistically significant extent, though perhaps I ought to have tested for this.

I implemented said labeling in a custom graph function, built on the pre-existing PolyGraphs function for generating a complete (fully-connected) network. This is not ideal— it means that the custom PolyGraphs Operation I wrote is only runnable on a complete network. Fortunately, this issue can be easily addressed.

The group labels are stored in a tensor with the same shape as the tensor for the network nodes. Each node lines up with its label across these tensors. There is, therefore, nothing that demands the labels be assigned to the nodes at the creation of the graph. A new tensor could be created inside of the custom operation, at the same shape as the tensor for the nodes. This would make the custom operation compatible with all pre-existing PolyGraphs functions for creating graphs of different types.

3.2. Custom PolyGraphs Operation

To simulate testimonial injustice, I needed a way to make some nodes discount the evidence (testimony) of some other nodes, all depending on the nodes' labels. The PolyGraphs Operation for simulating network polarization (OConnorWeatherallOp) does something similar, where nodes discount evidence they receive at a magnitude determined by the distance between the receiving node's belief and the testifying node's belief. The operation implements this evidence-discounting through Jeffrey's rule, which returns an updated belief for all nodes at each simulation step, taking not only each node's prior belief and incoming evidence, but each node's certainty regarding the evidence. So I scrapped the bit of the operation that calculated distance between belief, and inserted custom logic to model testimonial injustice.

I got the operation to work by defining a list of unique group labels. So, if there's two groups in the network, the list is [0, 1]. The list allows for iteration over each group in the network, which, in combination with masking the incoming evidence by senders of the current group (at that iteration), allows all incoming evidence to be received group-by-group. This meant that all the nodes in the network could discount the evidence of one particular group.

That is not exactly how I intended to model testimonial injustice, though. I didn't want *all* nodes to discount the evidence of nodes belonging to a certain group, I wanted only the nodes belonging to some other certain group to do the evidence discounting. To accomplish this, I used a pair of inverse masks. One mask is a tensor with the same shape as the group-label tensor, with a '0' if the corresponding node (by column) is not labeled with a '1', and a '1' if the corresponding node *is* labeled with a '1'. The other mask is that mask's binary inverse. I then took the first of those masks, and multiplied it by certainty in case of prejudice— a value I'd parameterized by hijacking the pre-existing 'mistrust' parameter. Doing this defines nodes in group 1 as those enacting testimonial injustice, or as I labeled them, 'prejudiced.' The other mask is left as is, and the two masks are summed together. Summing works here because the masks are each other's inverses— each mask's non-zero values overwrite the other's zero values. We end up with a tensor that has the specified certainty in case of prejudiced value for all the nodes belonging to group 1, and a '1' (no uncertainty) for all the nodes not belonging to group 1. To define nodes belonging to group 0 (it doesn't matter which group) as the victims of testimonial injustice, or, 'prejudiced-against,' I implement these masks if and only if the loop

is iterating over group 0 (and, consequently, the evidence sent by nodes belonging to group 0). For all other groups, a tensor of all ones is passed into the certainty parameter of Jeffrey's rule.

3.3. Running Simulations

To run simulations with the custom operation, I iterate over a predefined set of parameters using nested for-loops. The outermost loop iterates over different network sizes, and the subsequent loops iterate over the number of groups, values to plug in for certainty in case of prejudice, and a range of seeds. Each combination of these parameters, aside from 'seed', is a simulation configuration. I iterate over n seeds to run n simulations for each configuration.

Using nested for-loops instead of generating batch jobs for the Slurm workload manager results in a batch per simulation (each simulation is in its own batch). This is fine for CPU execution, but terribly un-optimized for GPU execution, where the overhead of initializing a batch would not be outweighed by the benefits of running batch size 1 in parallel. Given that I was indeed running the simulations on my CPU, this worked well.

3.4. Processing Results

To process the results, I loaded the simulation data into the PolyGraphs processor, and converted the processor into a Pandas dataframe for typical operations such as filtering, grouping, merging, averaging, etc. All of the statistical tests performed and most of the data-visualizations produced use some combination of these Pandas operations, Matplotlib/Seaborn, and Scipy/Numpy.

The exceptions to this statement are those visualizations that chart node belief over simulation step. Node beliefs are managed by the PolyGraphs processor, but I couldn't manage to filter the data directly within the processor. Although converting the data to a dataframe allows for easy filtering, it remains unclear whether or not the filtered dataframe can be converted back into a processor (probably not, I imagine going to a dataframe is lossy). Instead, I created a dictionary to store indices of simulations for each configuration, and populated the dictionary from a dataframe-version of the processor by filtering and appending. The PolyGraphs processor retrieves node beliefs for a specific simulation index using the '.beliefs' method, so I implemented a function (two functions that work in tandem) to take the indices for a simulation configuration from the dictionary, pass them into the processor, and return the average belief by group across all the given indices (later, I justify my decision to average node belief by group, by configuration).

4. Data Analysis

4.1. Impact of Evidence Order on Jeffrey's Rule

As described, each node in the network updates its belief using Jeffrey's rule. At each simulation step, all nodes first update based on evidence from group 0, followed by group 1, and so on. This arbitrary ordering does not affect belief-updating on evidence from any groups other than group 0, as the certainty is always set to 1 for those cases. And, when certainty is set to '1' for Jeffrey's rule, the formula is not sensitive to the order in which evidence is passed—Jeffrey's rule with certainty '1' is just Bayes' rule, which is not order-sensitive.

However, when updating based on evidence from group 0, the certainty values are not always set to 1.

In these instances, the order of evidence affects the updated belief returned by Jeffrey's rule. Given that the evidence order is set arbitrarily in the custom operation, I decided to test whether the network output showed a statistically significant difference compared to a network with a different evidence order.

To perform this test, I needed simulation data generated by an operation that updates nodes with a different ordering of groups. Therefore, I copied the custom operation I had already written, and added a line of code to shuffle (randomly rearrange) the list of groups in the network at every simulation step. I then ran this new operation with the same simulation configurations and seeds as the original.

With the new data, I was able to run a paired-samples test. Checking a patient's symptoms, giving that patient medicine, and then later re-checking that patient's symptoms makes for a typical example of such a test. The samples are paired by pre-intervention (the patient's symptoms before administering medicine) and post-intervention (the patient's symptoms after administering medicine), where the patient serves as their own control.

Because the network seed controls for all belief initialization (for a given size), outcomes of experiments (for a given size), and grouping of nodes (for a given number of groups and size), I can use each network seed in a manner analogous to the aforementioned patient—that is, as its own control. This is an idea I employ repeatedly across my analysis.

So, I defined the average steps taken by networks simulated with an arbitrary, *static* evidence-order at seed X as pre-intervention, and the average steps taken by networks simulated with an arbitrary, *shuffled* evidence-order at seed X as post-intervention. In doing so, I defined the shuffling of evidence-order as the intervention, which allowed me to test for whether the impact of evidence-order is statistically significant across all seeds.

Selecting an appropriate statistical test was straightforward. After confirming that the data was not normally distributed, and knowing that I didn't have enough data to take samples large enough for the distribution of those sample means to approach normality (Central Limit Theorem⁷), I had 2 choices, as can be seen in the following flowchart:

⁷ D. Montgomery and Runger, *Applied statistics and probability for engineers*.

Choosing a statistical test

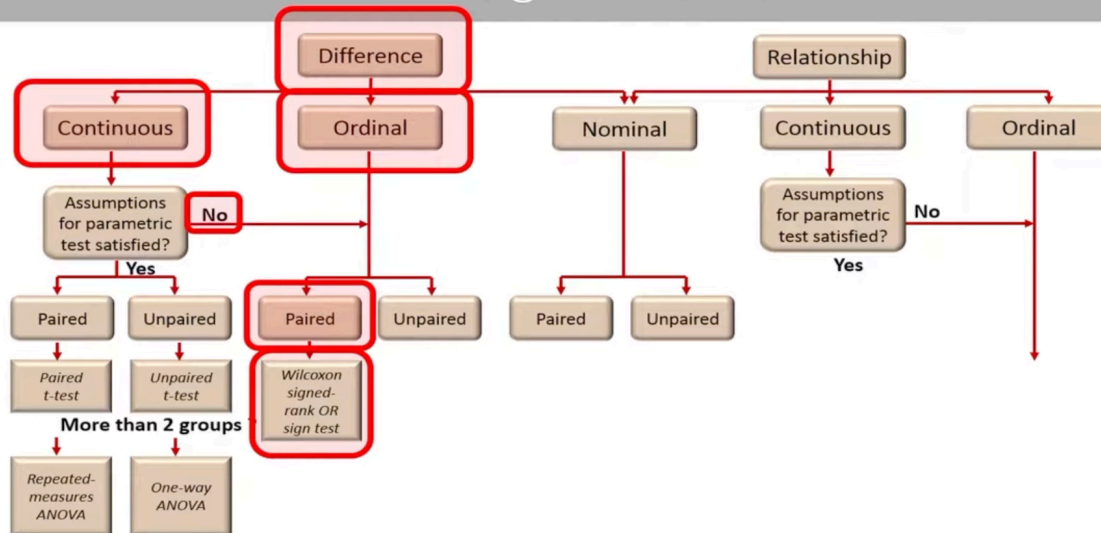


Figure 1. Choosing a statistical test⁸

Although the data at hand (steps) was continuous, it failed assumptions of normality, so I attempted a Wilcoxon signed-rank test. However, the data, too, failed the assumption of the Wilcoxon signed-rank test, in that the distribution of the differences between steps taken pre and post-intervention was not symmetrical. Therefore, I ran a sign-test.

4.1.1. Sign Test: Change in Steps by Evidence-Order

Intervention: Shuffling the order of evidence (which group has their evidence updated on first)

Null Hypothesis: Shuffling the order of evidence, as applied in Jeffreys' Rule, does not result in a statistically significant difference in the average number of steps taken by the networks

Alternative Hypothesis: Shuffling the order of evidence, as applied in Jeffreys' Rule, *does* result in a statistically significant difference in the average number of steps taken by the networks (two-sided, as in the difference could be due to the post-intervention value being greater and lesser)

Test Results: The sign test returns a p-value of 1.0, indicating that there is no evidence to reject the null hypothesis. In other words, the test confirms that no statistically significant difference emerges in the steps values after shuffling the order of evidence. Although the average steps across all simulation configurations are different from pre to post at 59 of the 100 seeds tested, those 59 differences are composed of a fairly even amount of positive and negative values.

4.2. Network-Wide Analysis

Having established that evidence order is not distorting my simulation results, I proceeded to compare the simulation configurations and visualize the relevant data. This analysis focuses on a network-wide perspective, rather than examining node-level beliefs or average group beliefs. It provides a macro-level overview, focusing on differences between whole networks across the simulation

⁸ A. Tilevik, *How to choose an appropriate statistical test*.

configuration space.

4.2.1. Distributions of Steps

We can graph the distributions of steps taken by the network across different configurations to get an idea of the variability and central tendency of steps for each simulation configuration. Box plots are a good choice for visualizing continuous data like this, where a lot of information would be lost in, say, visualizing the average for each configuration in a bar chart.

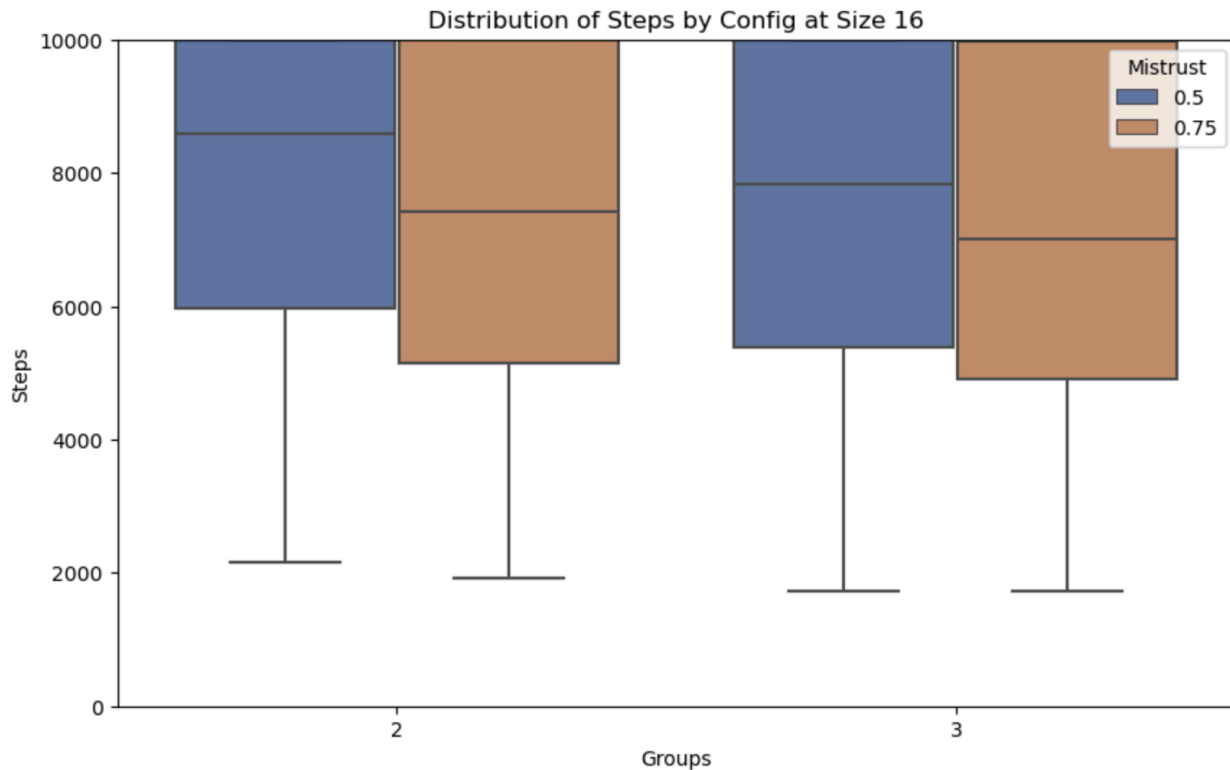


Figure 2. Distribution of steps taken for all configurations at network size 16. Each box plot displays the number of steps taken (until convergence or step 10,000) for each of the 100 simulations at that configuration.

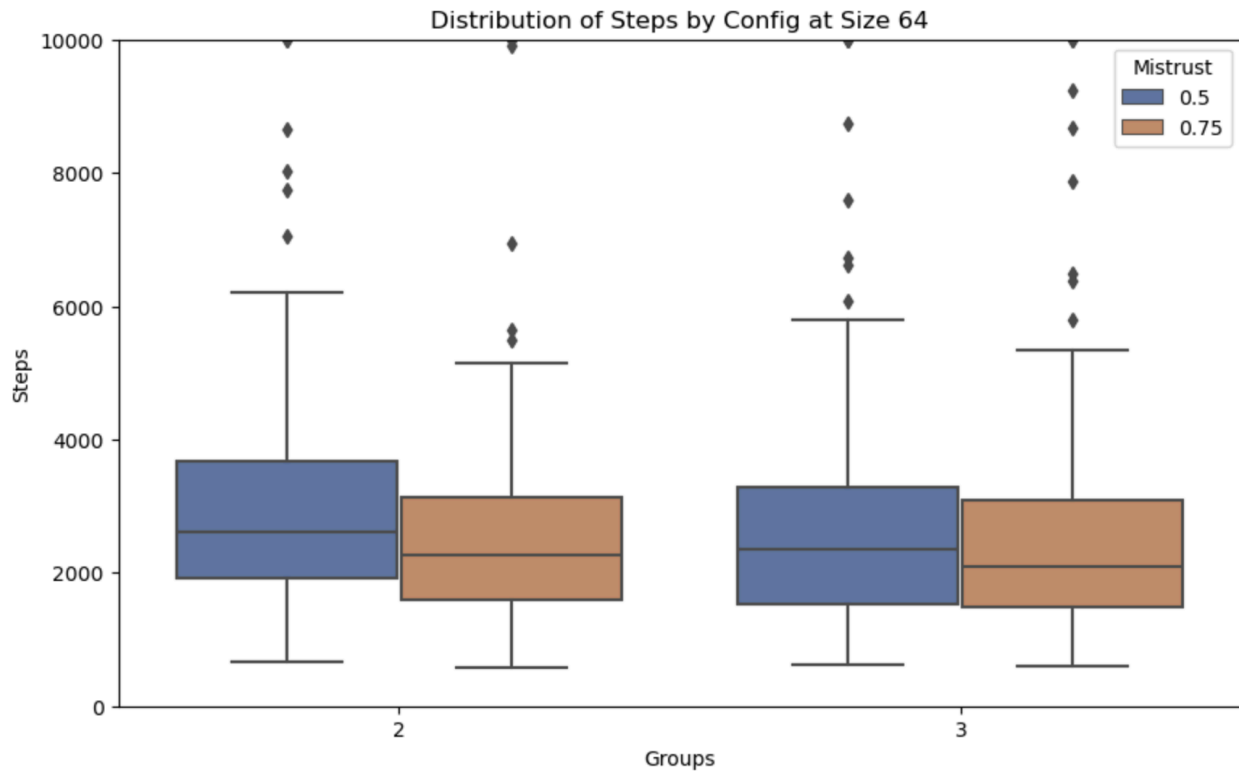


Figure 3. Distribution of steps taken for all configurations at network size 64. Each box plot displays the number of steps taken (until convergence or step 10,000) for each of the 100 simulations at that configuration.

At all configurations, across both network sizes, central tendency is lower when there is less prejudice in the system. This is most immediately apparent in considering steps (y-axis) by certainty in case of prejudice (hue) at each group pairing. When there is a higher magnitude of prejudice (certainty at 0.5 in case of prejudice), the network seems to take longer (more steps), at least in terms of median steps taken. This is also true when we compare configurations at the same ‘certainty in case of prejudice’ (hue) across ‘number of groups’. This makes sense, in considering that the groups are always of equal size— when there are three groups, the nodes that enact testimonial injustice account for just one third of the network, as opposed to the half they account for when there are only two groups.

At size 64, we also see that the upper whiskers extend to a greater ‘steps’ value when there is more prejudice in the network. As before, this holds true across the aforementioned comparisons, aside from one case, where, for an unidentified reason, the upper whisker for *simulation-configuration: size 64, 3 groups, 0.75 certainty in case of prejudice* extends to a greater value than the upper whisker for *simulation-configuration: size 64, 2 groups, 0.75 certainty*. Note that the effect is only visible at size 64 because the networks at size 16 are significantly slower— to such an extent that the steps distribution is visibly cut-off by the maximum steps limit of 10,000.

4.2.2. Network-Action Proportions

In this section, we analyze the distribution of network-action proportions across different simulation configurations. By visualizing the percentage of how many simulations converged to truth per network, we gain insight into how the network success rate (success is converging to the truth) varies with changes in network parameters. I visualize two proportions here— the number of networks that converged to the truth at configuration X divided by all networks at configuration X, as well as the

number of networks that converged to the truth at configuration X divided by only those networks that converged at configuration X (networks that converged to truth + networks that converged to falsehood). This is in hopes of mitigating any false data interpretation that might result from the 10,000 step maximum, where some simulations don't have enough "time" to converge one way or the other, and are instead left with an undefined network action.

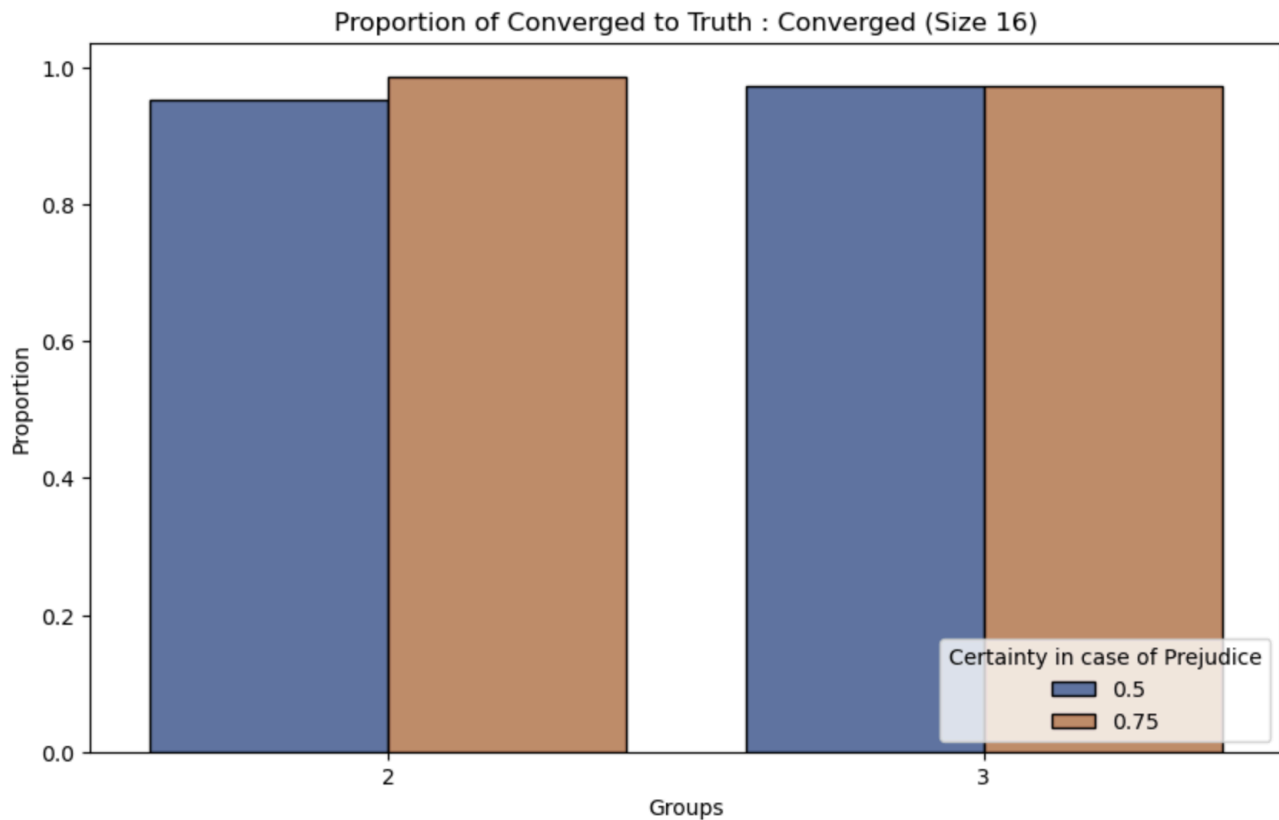


Figure 4. The number of simulations that converged to truth, divided by the number of simulations that converged to truth or falsehood, for all simulation configurations at size 16.

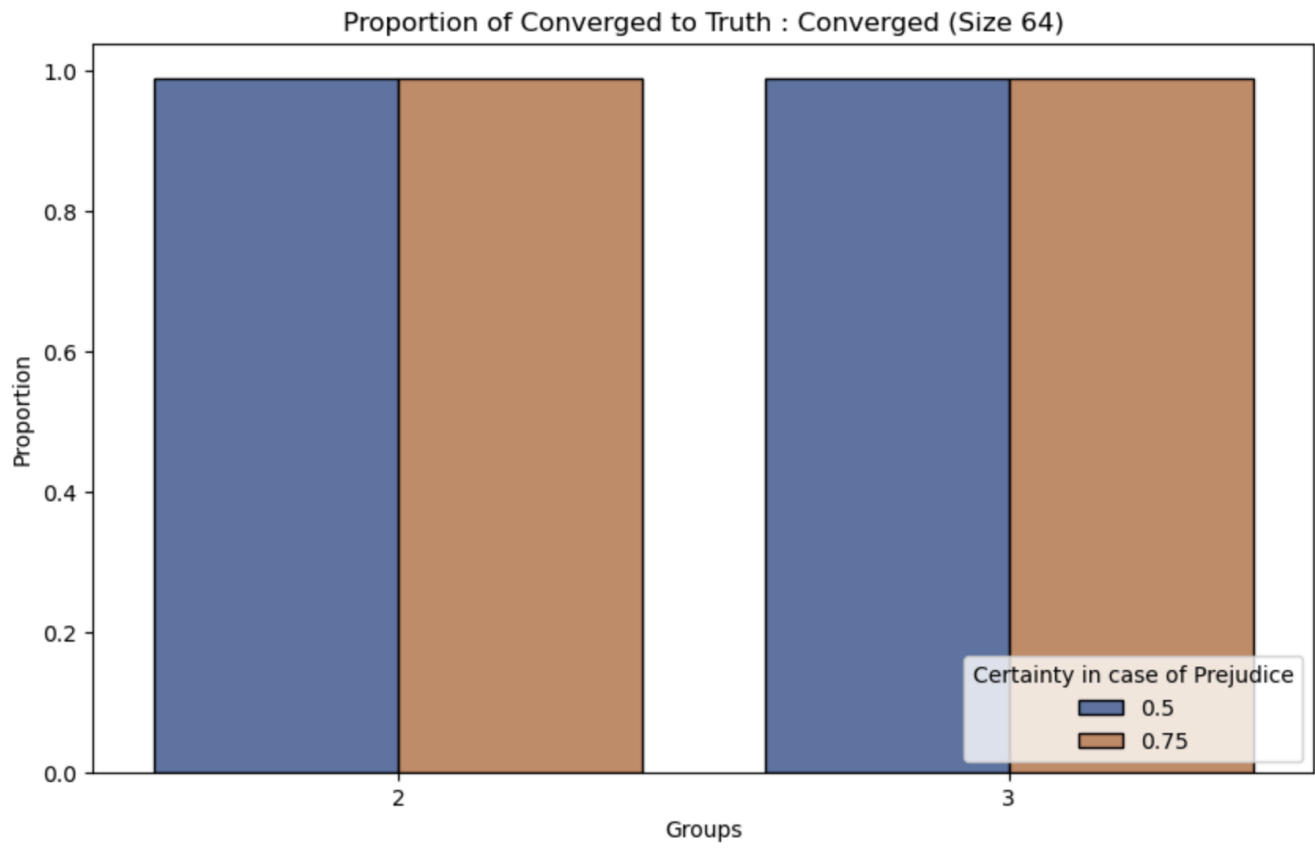


Figure 5. The number of simulations that converged to truth, divided by the number of simulations that converged to truth or falsehood, for all simulation configurations at size 64.

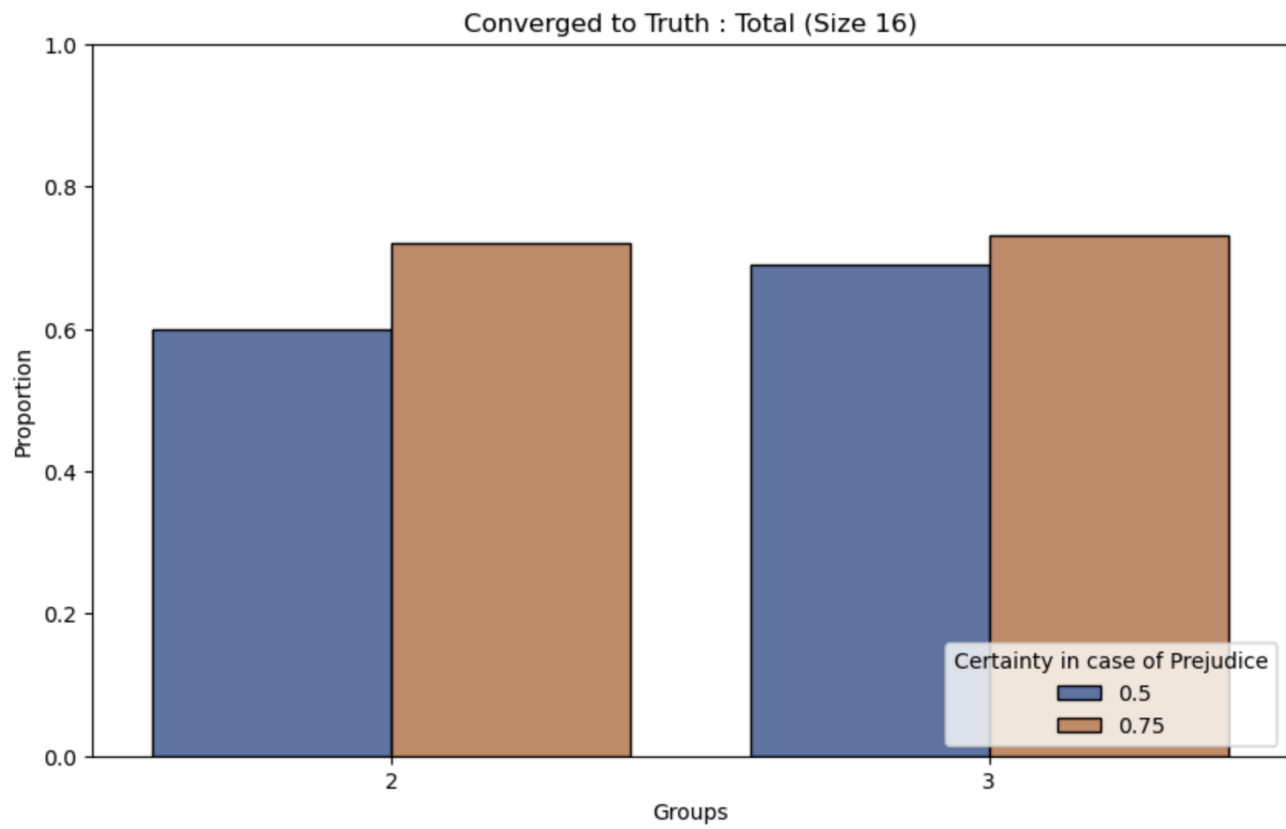


Figure 6. The number of simulations that converged to truth, divided by the total number of simulations for all simulation configurations at size 16.

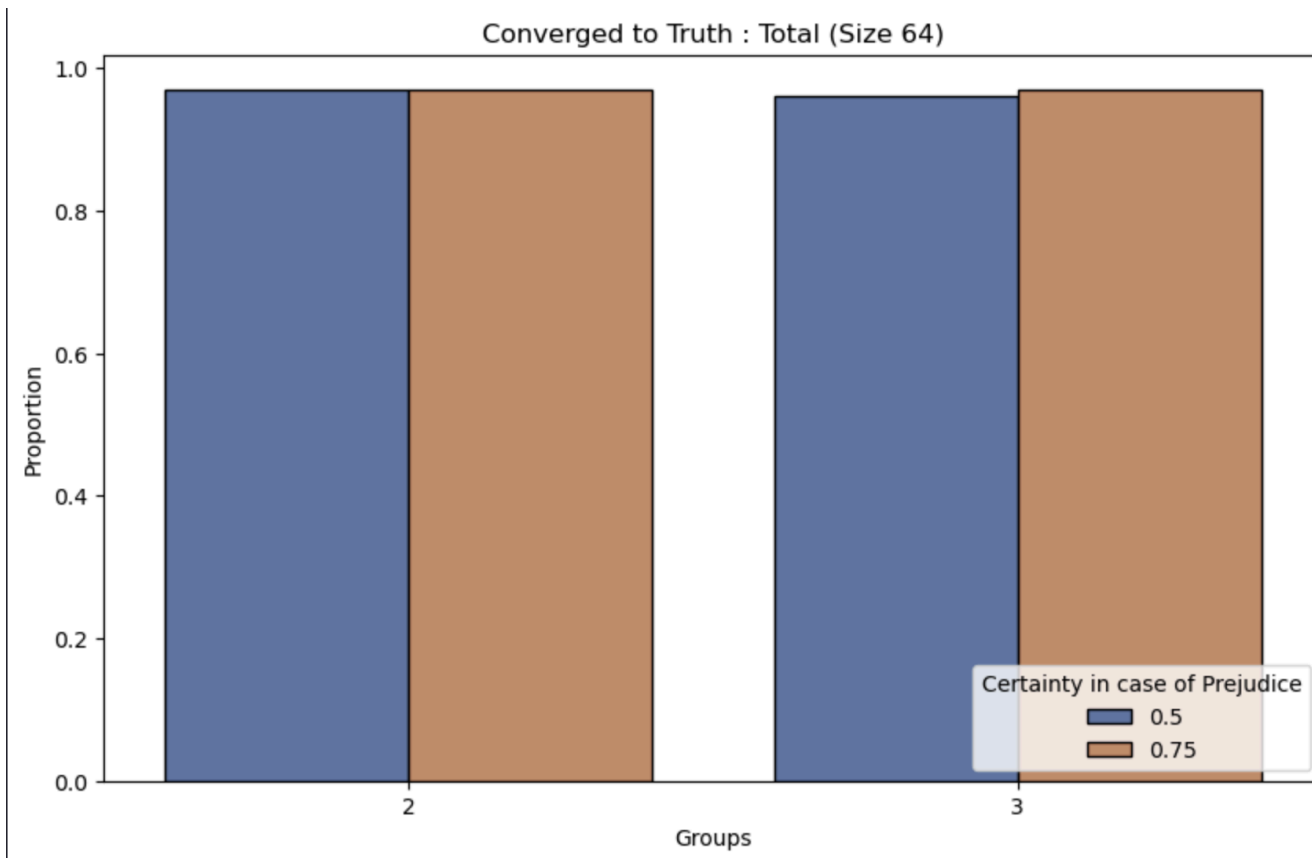


Figure 7. The number of simulations that converged to truth, divided by the total number of simulations for all simulation configurations at size 64.

The bar charts do not indicate a great amount of difference in proportion over the network parameter space, aside from Figure 6. This checks out with what we saw displayed in Figure 2— by having seen the distributions of steps, we already know that a lot of simulations were cut short at every size 16 network configuration. It makes sense, then, that at size 16, when we divide the number of simulations that converged to the truth by the total number of simulations, we get much lower values. The high number of simulations that needed more than 10,000 steps to converge is “watering-down” the “success-rate.”

It holds true across all the charts, that, when there is a disparity in proportion by ‘certainty in case of prejudice,’ it is the simulation configuration with the higher magnitude of prejudice (0.5 CIP) that has the lesser proportion of simulations that converge to truth. Note that, in comparing configurations at 0.5 CIP by number of groups, Figure 4 indicates an expected disparity— simulation configuration 16_2_5 has a lesser proportion (converged) than configuration 16_3_5. Figure 7, however, indicates the inverse— simulation configuration 64_2_5 has a greater proportion (total) than 64_3_5! I cannot yet explain why this is happening, but it ought not be forgotten.

4.2.3. McNemar’s Test: Impact of CIP on Convergence to Truth

As mentioned in the previous section, it appears that ‘certainty in case of prejudice’ has an effect on the success rate (convergence to truth), but not by much. Fortunately, we don’t need to rely on sight to determine whether the visualized disparities are statistically significant— we can, again, test for that.

As before, I performed a paired-samples test, but this time, the data at hand is different. Previously, I dealt with paired samples of non-normally distributed, continuous data. Here, we have non-normally distributed paired samples, but the data is nominal: 'A' for 'converged to falsehood,' 'B' for 'converged to truth,' and '?' for 'did not converge before the step limit.' It seems simple, then—I must merely choose a test for non-normally distributed paired samples of nominal data. Unfortunately, however, with the inclusion of the unconverged networks, there are three possible outcomes for each simulation. This is unfortunate because there is no test for paired, nominal data where the dependent variable has greater than two possible outcomes—at least, not one that I could find.

The two best fits are McNemar's test and the chi-square test. Both, however, are incompatible with the data at hand. McNemar's test doesn't quite work because it assumes a dichotomous dependent variable (two possible outcomes).⁹ As explained, I need here to test for a *trichotomous* dependent variable. The chi-square test can handle a trichotomous dependent variable, but disregards sample-pairs.¹⁰ So, technically, I could use either test, either by mapping the trichotomous dependent variable to a binary, or by making content without sample-pairs. Of these two cons, the former seems less unappealing. Disregarding sample-pairs means failing to control for initial conditions, as initial conditions are controlled by pairing pre and post-intervention values by seed. In mapping the dependent variable to a binary, I squash the 'A' and '?' outcomes into one outcome, '¬B', and so it is reductive, but at least it allows me to control for initial conditions. Thus, I performed McNemar's test on the data, asking, 'Does certainty in case of prejudice affect a network's success rate to a statistically significant extent?' (where meeting the 10,000 step limit is considered failure).

Intervention: Decreasing certainty in case of prejudice from 0.75 to 0.5 (increasing the severity of testimonial injustice)

Null Hypothesis: Decreasing certainty in case of prejudice does not result in a statistically significant change in network action, where a network can either converge to truth or not converge to truth

Alternative Hypothesis: Decreasing certainty in case of prejudice results in a statistically significant change in network action, where a network can either converge to truth or not converge to truth

McNemar's Test Results			
	Config	Statistic	p-value
0	16_2	0.00	0.00049
1	16_3	1.00	0.22
2	64_2	1.00	1.0
3	64_3	1.00	1.0

Figure 8. Results of McNemar's Test, indicating that the null hypothesis holds for all configurations aside from 'size: 16, groups: 2'.

I've colored the rows of the table in Figure 8 red when we cannot, with great confidence, reject the null hypothesis, as determined by the p-value. So, the test results indicate that it would be very

⁹ *Ibid.*

¹⁰ *Ibid.*

unlikely for us to be observing this difference in network action pre and post-intervention at simulation configuration 16_2 if the null hypothesis were true. In other words, we have strong evidence to reject the null hypothesis at that configuration.

The results provide no evidence to reject the null hypothesis for simulation configurations 64_2 and 64_3. This checks out— we know from Figures 5, 7 that pretty much all of the simulations at these configurations converge to truth, regardless of whether certainty in case of prejudice is 0.75 or 0.5. The results are less clear for configuration 16_3. The p-value indicates that we could expect to see this difference in network action pre and post-intervention 22% of the time, assuming the null hypothesis. That's not very probable, but it is not improbable enough to reject the null hypothesis at the configuration with great confidence.

The results seem to suggest that there is a threshold on severity of testimonial injustice in the network, which, once crossed, flips the network's outcome from convergence to truth, to convergence to falsehood. This, however, is likely a misinterpretation. Remember that, to make the dependent variable compatible with McNemar's test, I reduced my trichotomous dependent variable to a binary. It seems more reasonable, then, to posit that the affected simulations at configuration 16_2 are slowed down to the point of reaching the 10,000 step limit. This is unfortunate, as we are not looking for network speed, but network reliability. In retrospect, it may have been better to run a chi-square test here, even in its abandonment of controlling initialization by way of seed-pairing. Combining measures of network speed and reliability into a single metric renders the results mostly uninterpretable, particularly in considering the Zollman Effect— which describes network reliability and network speed as mutually exclusive in terms of optimization.

4.3. Initial Group-Dynamics & Node-Belief Analysis

Having analyzed and tested general network results, I shifted into an investigation of group dynamics and node belief— that is, relationships in node-belief by group. Perhaps, down here, we can find insight into how prejudice interacts with individual belief and group belief, beyond its effects on the whole network. At this point, I expected to find that prejudiced nodes are, on average, worse off (epistemically) than their peers.

4.3.1. Average *Average-Belief-by-Group* by Simulation Configuration

In order to visualize node belief by group over time (steps), representative of each simulation configuration, I was confronted with two options:

1. Take a small, random sample from each simulation configuration, and compute average belief by group for the individual simulations that were selected
2. Calculate the average belief for each group at every simulation. Then, average the resulting group-averages by simulation configuration to obtain an overall average for each group by simulation configuration.

I went with the second option, due to my consideration of what I call, 'The Incidental-Filtering-by-Prejudice Effect,' but which I am not sure is any different from 'The A-Broken-Clock-is-Right-Twice-a-Day-Effect.' See, in taking a sample of individual simulations, we run more risk of presenting misleading data, where the smallness of the sample size is proportional to the magnitude of the risk. In fact, I tried visualizing a small sample (the sample can't be too big, because

then we can't compare across the many charts), and was met by the aforementioned effect. Take, for example, the following simulation at configuration size 64, 3 groups, 0.5 certainty-in-case-of-prejudice:

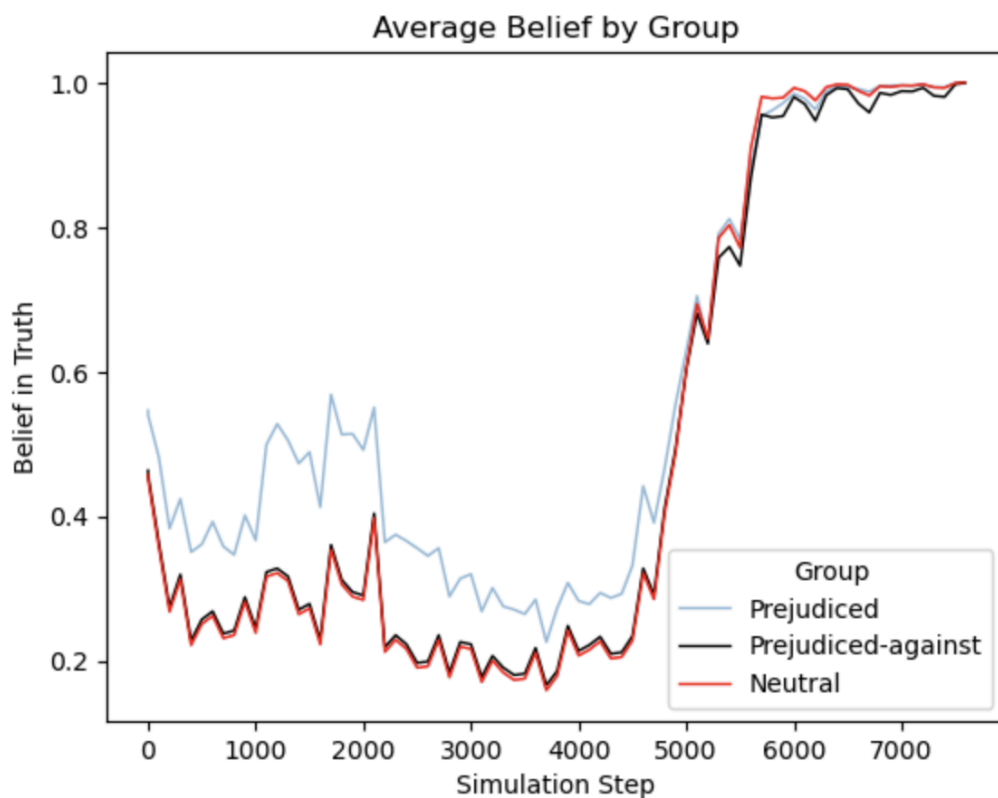


Figure 9. A line chart displaying average belief by group for a single simulation, indicating that nodes in the prejudiced group are better off, epistemically, on average, than the nodes in the non-prejudiced groups

Here, we see that the nodes in the prejudiced group are better off (epistemically, on average) than the nodes in the non-prejudiced groups. I identify two potential reasons for this effect:

1. The nodes in the prejudiced group initialized with higher average belief in truth than the other groups (we can see that this is the case)
2. The nodes in the prejudiced-against group so happened to be disproportionately responsible for a streak of misleading evidence (flipping a coin ten times, and getting heads every time)

Reason 1 is self-explanatory, but to understand reason 2, we must remember that nodes in the prejudiced group discount evidence from nodes in the prejudiced-against group. It follows, then, that the nodes in the prejudiced group would see an increase in belief in truth if the nodes that they are prejudiced against were to send a lot of evidence that suggested the opposite of the truth.

My choice, then, is prophylactic. By averaging the averages in the manner I have described, I preemptively address the two reasons listed—the impact of any particular belief-initialization is lost to the uniform distribution of belief-initialization across all the simulations at a given configuration. Similarly, the impact of any unlucky streak of misleading evidence is lost to the overall distribution of evidence (just as we know that flipping a coin ten times and getting heads every time is distributionally unlikely). So, I expected The Incidental-Filtering-by-Prejudice Effect to disappear upon

‘averaging the averages’ across all the seeds of a given simulation configuration.

There is one more point to address in regards to averaging group-averages from multiple simulations— simulations are of varying lengths. This is a problem because node-belief exists over time (simulation steps). So, for example, if one of the simulations ends at step 5,000 by converging, that simulation’s average-belief-by-group data will not be considered in calculating the overall average at the next simulation step. I figured that this would be fine, insofar as I refrain from comparing the resulting group averages across simulation configurations. In other words, the processed data should still be okay for accurately comparing groups within a configuration— the magnitude of the y-axis becomes arbitrary, and we instead look for disparity in y-axis by group. This is because, if, say, a simulation ends early, the resulting inaccuracy in the line chart would be equal for all groups: the only way for a simulation to end early is to converge.

I re-address this later in an attempt to compare average average-belief-by-group by simulation configuration *across* simulation configurations in a later section.

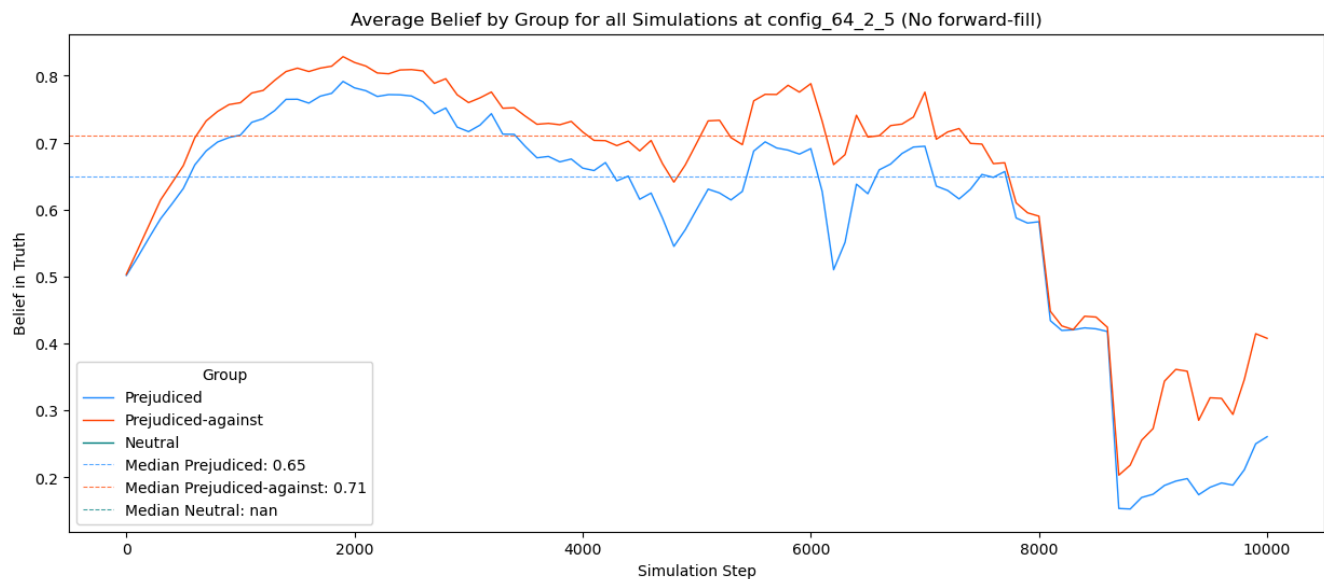


Figure 10. A line chart of average belief in truth by group for 100 simulations at size 64, 2 groups, 0.5 certainty in case of prejudice

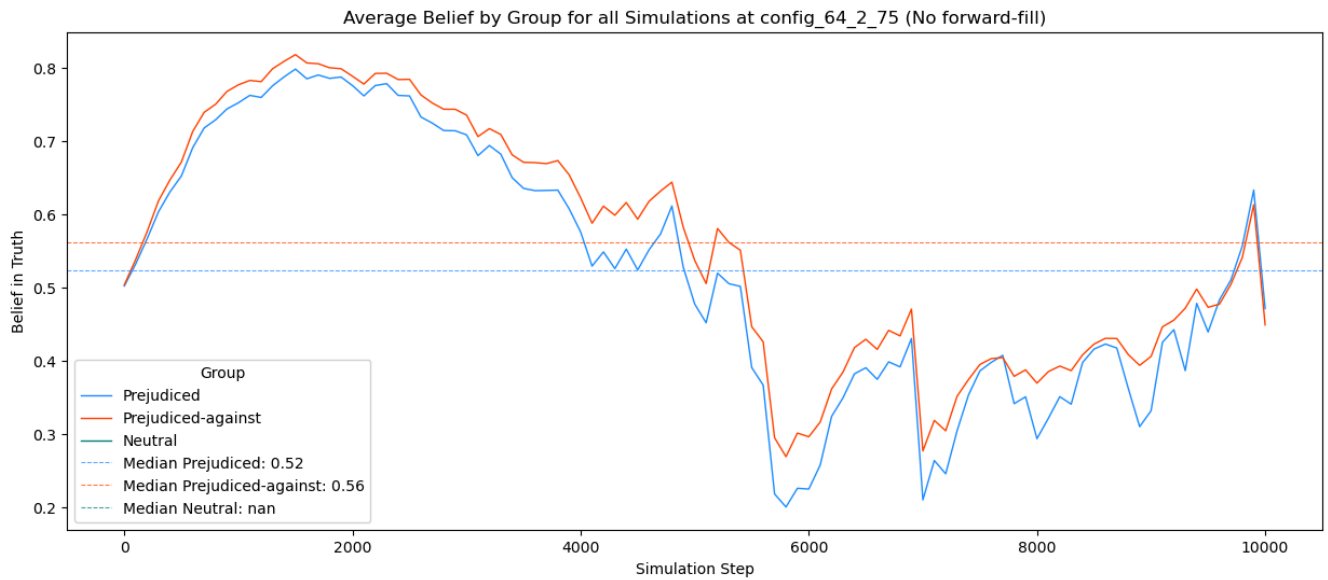


Figure 11. A line chart of average belief in truth by group for 100 simulations at size 64, 2 groups, 0.75 certainty in case of prejudice

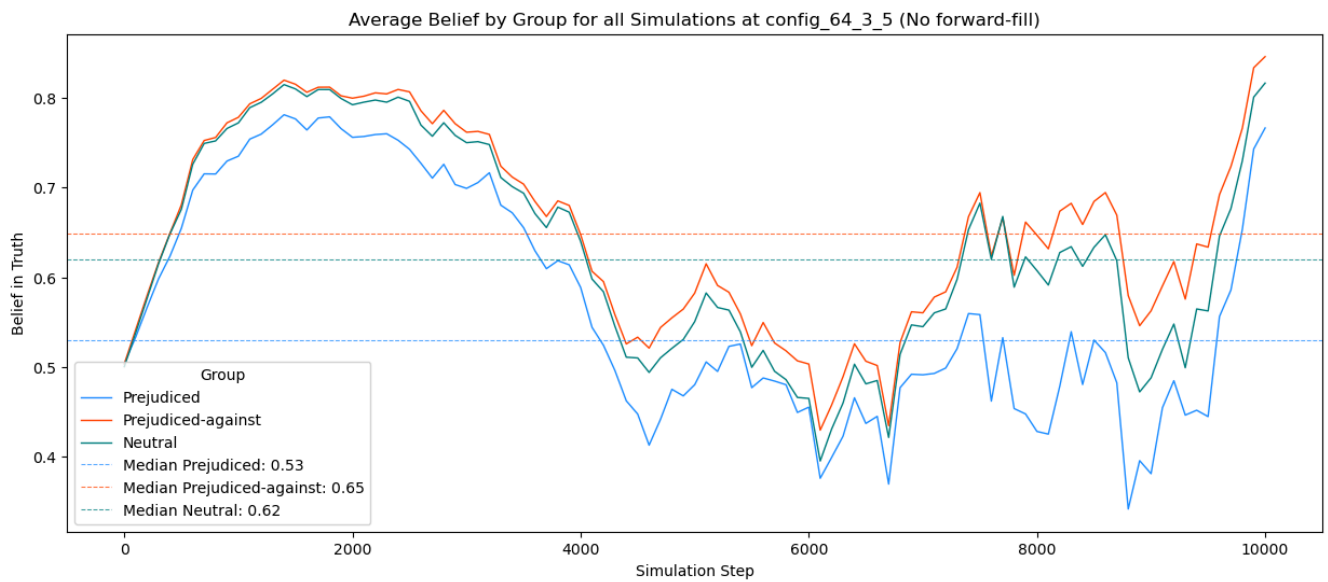


Figure 12. A line chart of average belief in truth by group for 100 simulations at size 64, 3 groups, 0.5 certainty in case of prejudice

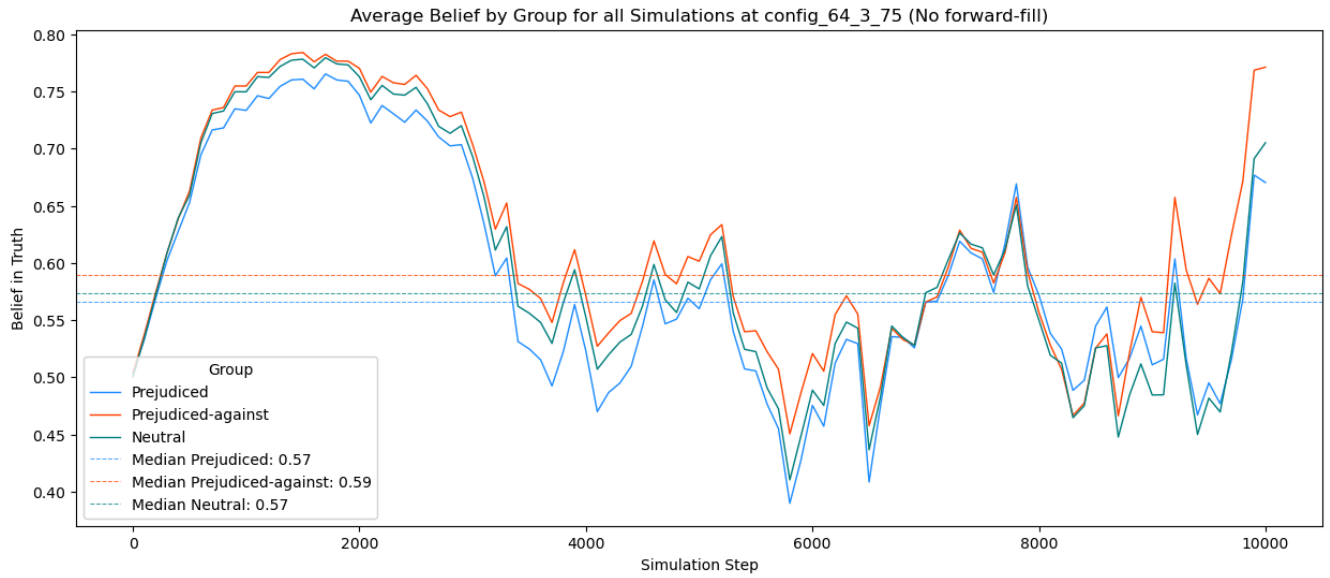


Figure 13. A line chart of average belief in truth by group for 100 simulations at size 64, 3 groups, 0.75 certainty in case of prejudice

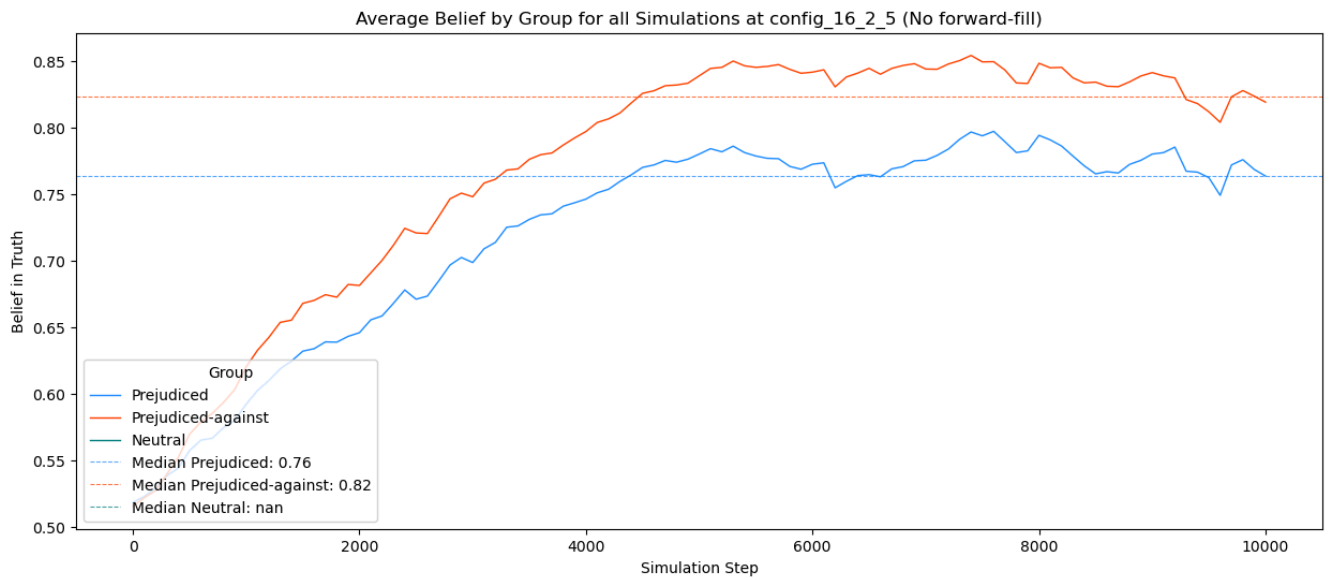


Figure 14. A line chart of average belief in truth by group for 100 simulations at size 16, 2 groups, 0.5 certainty in case of prejudice

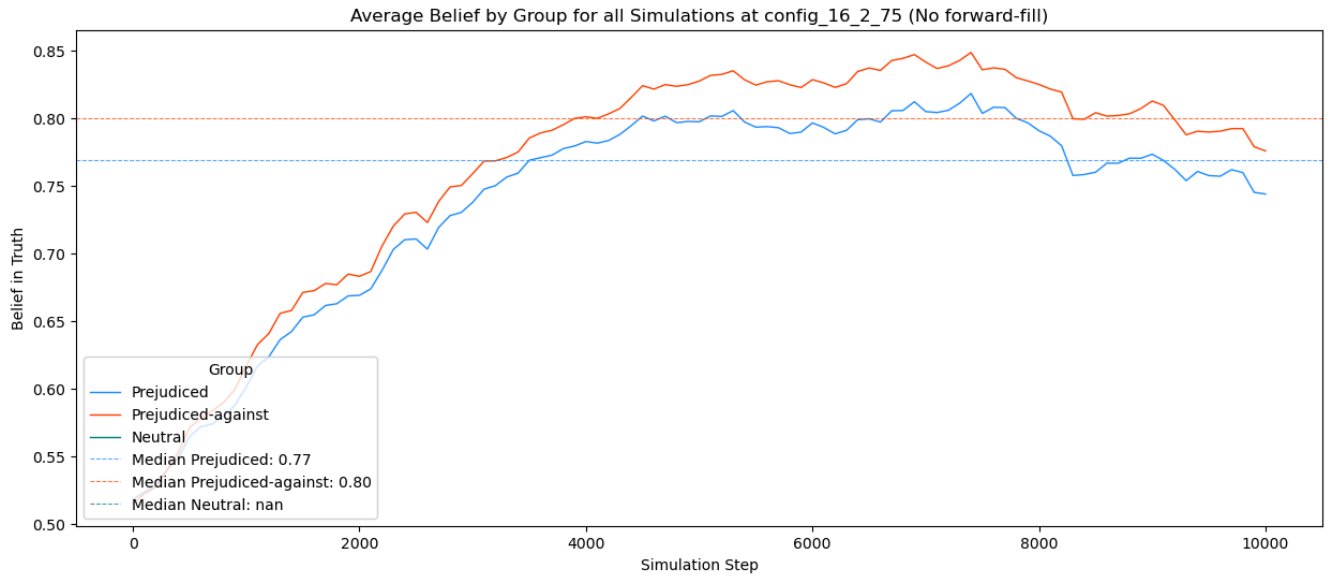


Figure 15. A line chart of average belief in truth by group for 100 simulations at size 16, 2 groups, 0.75 certainty in case of prejudice

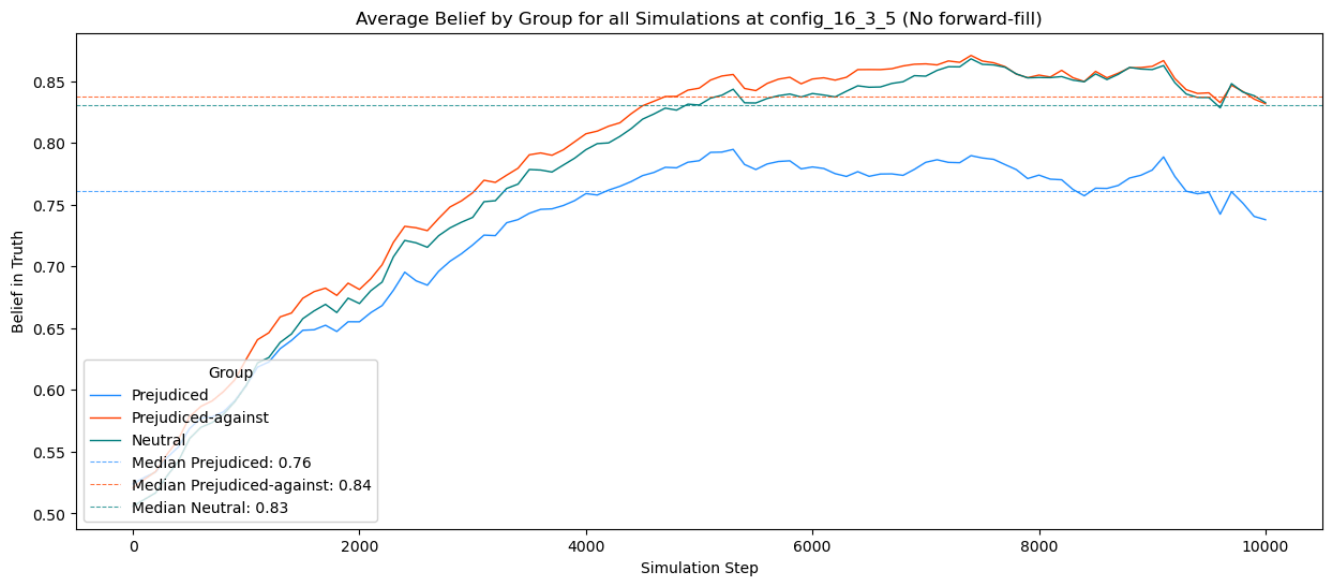


Figure 16. A line chart of average belief in truth by group for 100 simulations at size 16, 3 groups, 0.5 certainty in case of prejudice

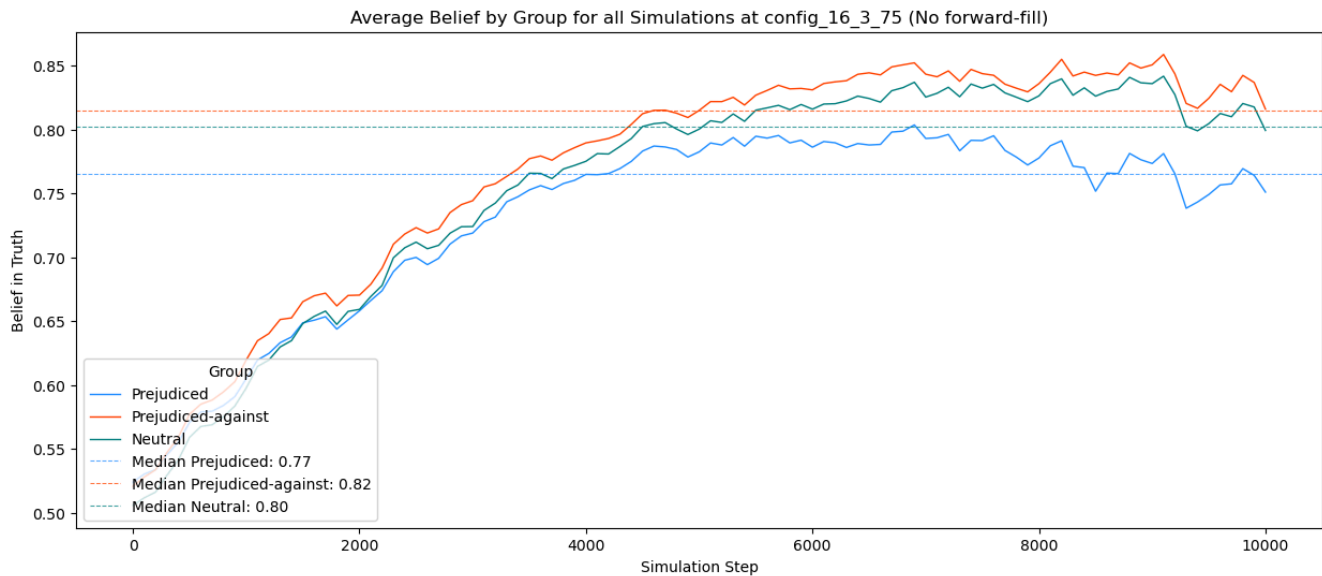


Figure 17. A line chart of average belief in truth by group for 100 simulations at size 16, 3 groups, 0.75 certainty in case of prejudice

For all simulation configurations, the nodes in the prejudiced group are clearly worse off, epistemically, than nodes in the non-prejudiced groups. This seems reasonable— nodes in the prejudice group update on evidence less than nodes in the other groups, and all the evidence is good evidence. It is strange, however, that, at simulation configurations with a third group, the neutral nodes also seem to be consistently, epistemically worse off than nodes in the prejudiced-against group. Unlike as is the case for nodes in the prejudiced group, I cannot point to a mechanism which might explain why nodes in the neutral group deviate so significantly from nodes in the prejudiced-against group. The charts seem to suggest that, somehow, by having their evidence discounted in instances of testimonial injustice, nodes in the prejudiced-against group see an increase in their average closeness to truth. I say this because the custom operation treats nodes in the prejudiced-against and neutral groups exactly the same, aside from that one difference— that nodes in the prejudiced-against group have their evidence discounted.

Initially, I ran a sign test for each simulation configuration to confirm that nodes in both the prejudiced and neutral groups were worse off than the nodes in the prejudiced-against group. And, sure enough, the sign test indicated that this was indeed true at every simulation configuration, for both groups. I have, however, decided to exclude these tests:

To properly test for the group-dynamics in question, I needed separate data. It is bad practice to test a hypothesis regarding the existence of some pattern on the same data that was used to generate said hypothesis. That is why, typically, one splits the total data, and keeps blind from one part for testing. I did not do this, initially. Fortunately, the data is all synthetic— to get more of it, one need only run more simulations. So that is what I did.

4.4. Interim Group-Dynamics & Node-Belief Analysis

I ran 100 more simulations at each of the eight configurations, extending the limit on simulation steps from 10,000 to 20,000 steps. In addition to this, I ran a control set by running 100 simulations at each of four size and group combinations, where certainty in case of prejudice was set to 1 (no uncertainty,

no testimonial injustice). There are half the amount of control configurations as experiment configurations— for example, experiment configuration 64_2_5 and experiment configuration 64_2_75 share the control configuration, 64_2_1.

In order to ensure an accurate comparison between any experiment configuration and its control, I re-addressed the problem described in section 4.4.1 (averaging simulations of varying lengths). This time, when averaging the group-averages of an individual simulation with the group averages of all the other individual simulations at that same configuration, I implemented forward-fill. This means that the final value for a given simulation that converged before step 20,000 will be repeated (and taken into account when calculating the average) across all steps until step 20,000.

Note that the following charts are on a logarithmic scale, to increase readability given that there are now 6 lines on each chart (3 groups in the experiment, 3 groups in the control).

4.4.1. Average Average-Belief-by-Group by Simulation Configuration, Again

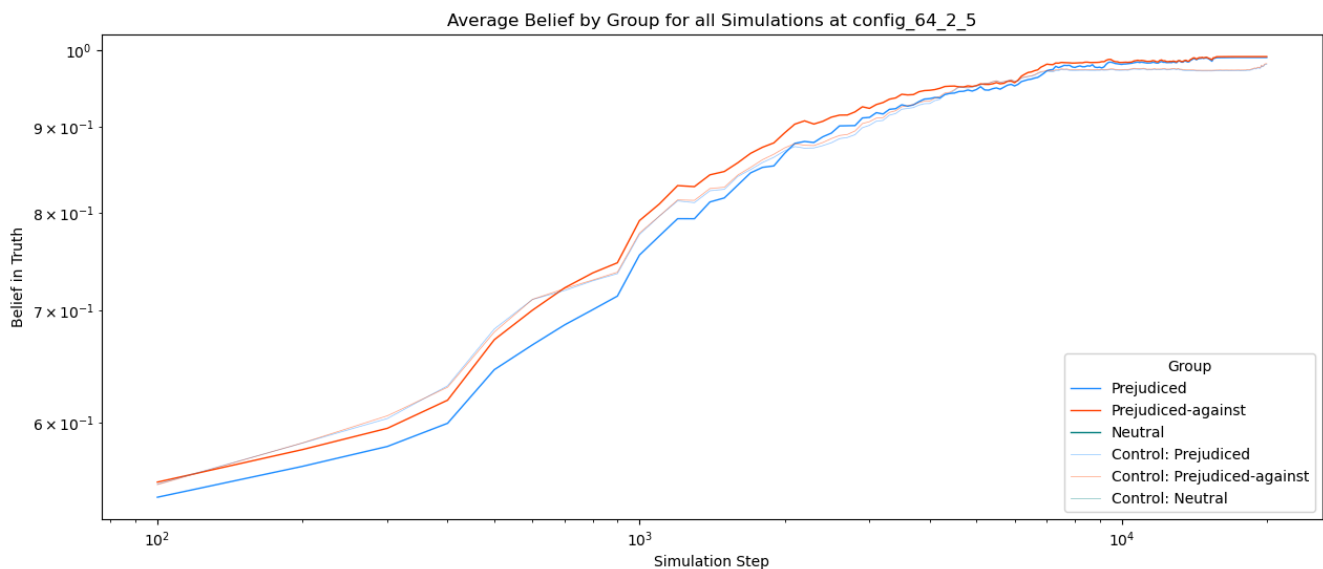


Figure 18. A line chart of average belief in truth by group for 100 simulations at size 64, 2 groups, 0.5 certainty in case of prejudice, alongside its control

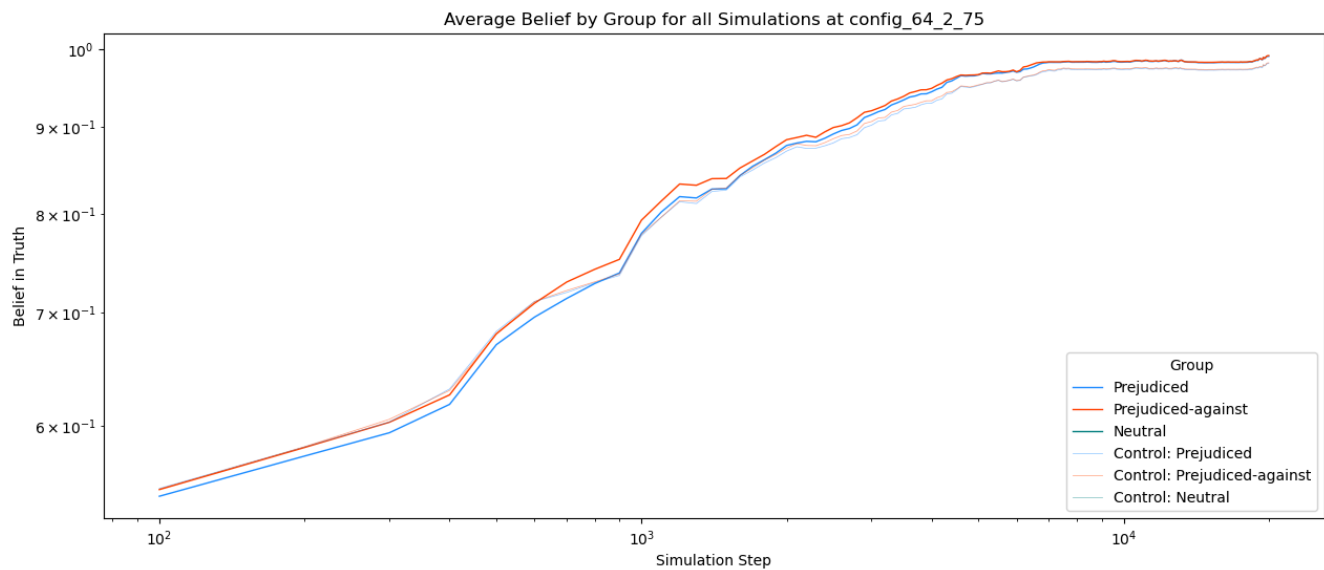


Figure 19. A line chart of average belief in truth by group for 100 simulations at size 64, 2 groups, 0.75 certainty in case of prejudice, alongside its control

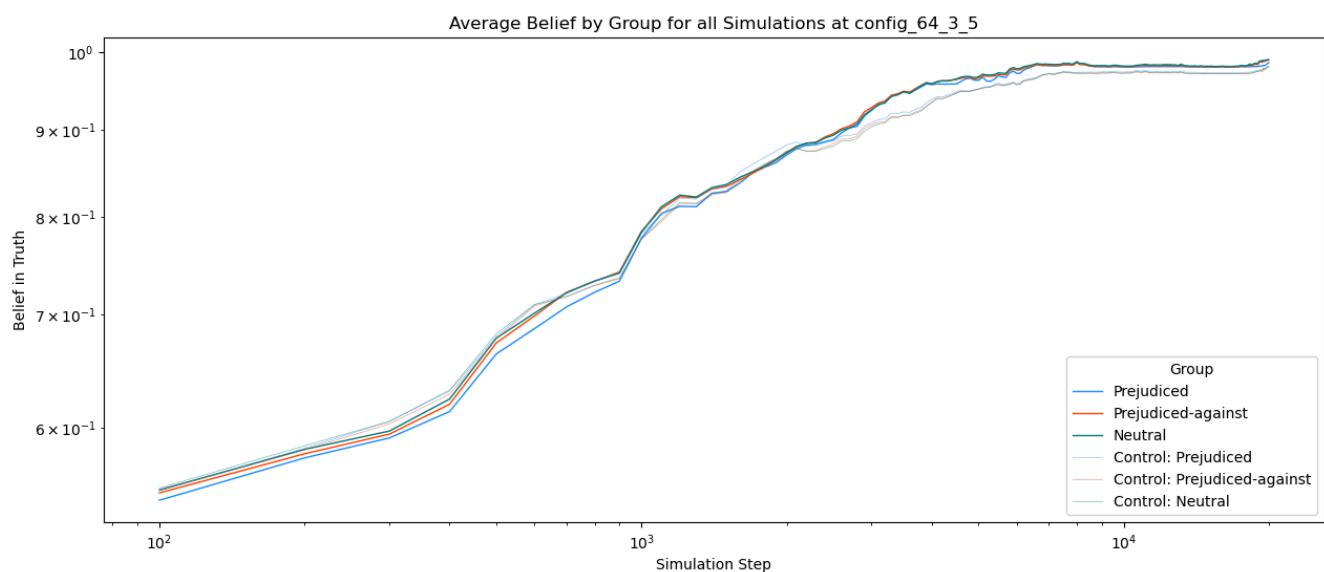


Figure 20. A line chart of average belief in truth by group for 100 simulations at size 64, 3 groups, 0.5 certainty in case of prejudice, alongside its control

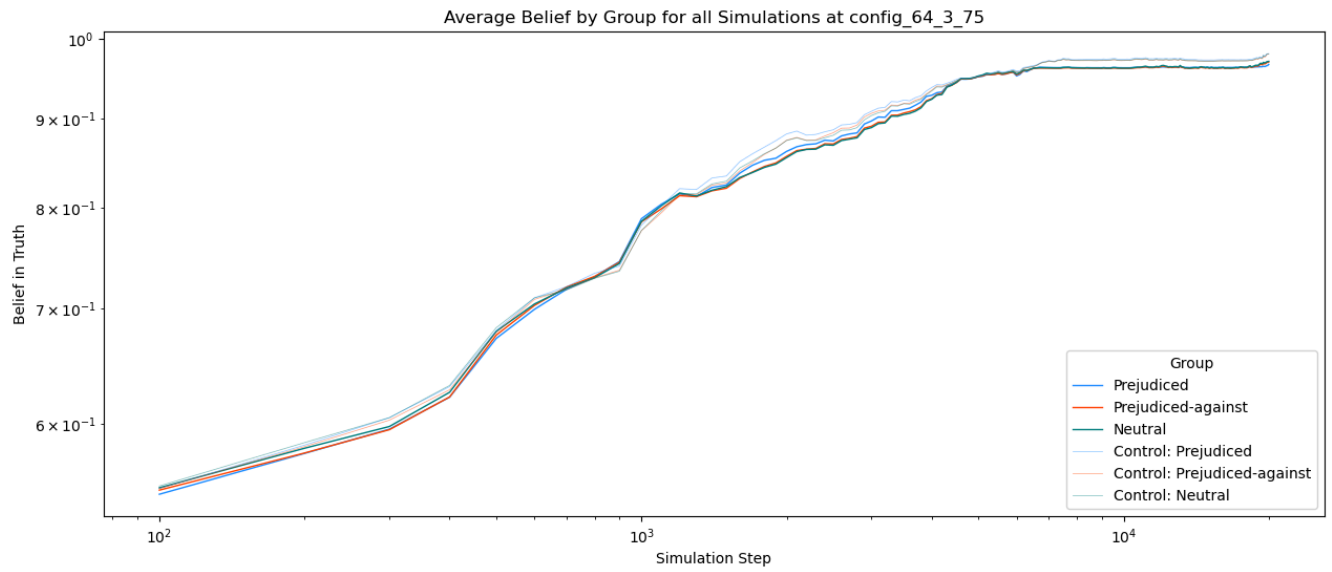


Figure 21. A line chart of average belief in truth by group for 100 simulations at size 64, 3 groups, 0.75 certainty in case of prejudice, alongside its control

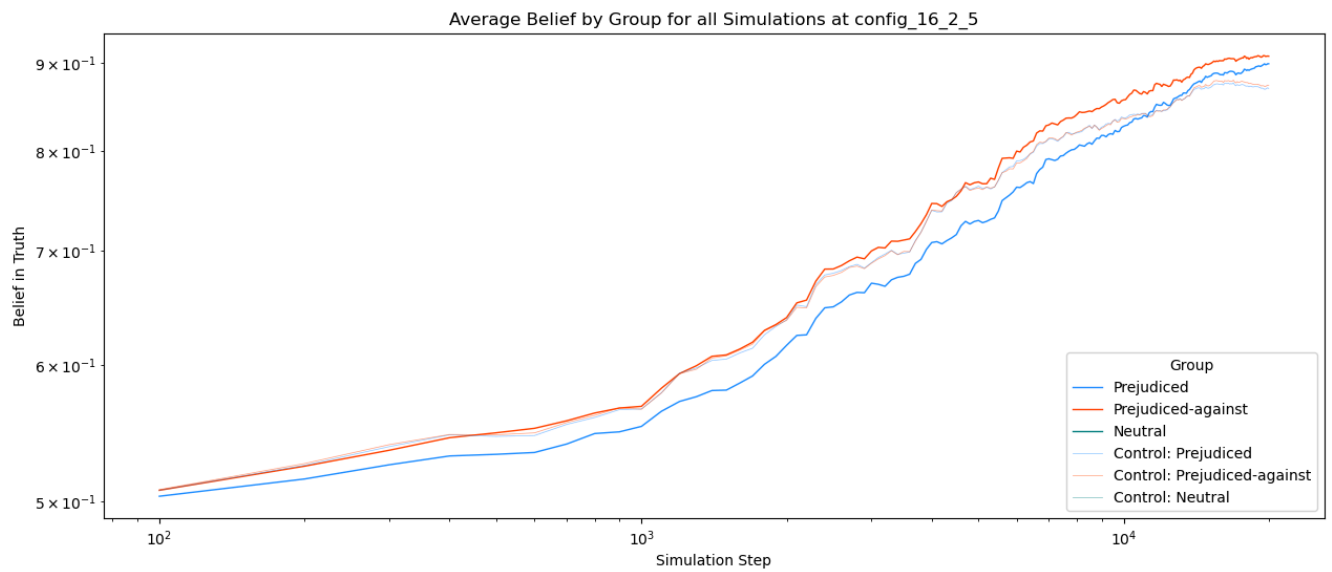


Figure 22. A line chart of average belief in truth by group for 100 simulations at size 16, 2 groups, 0.5 certainty in case of prejudice, alongside its control

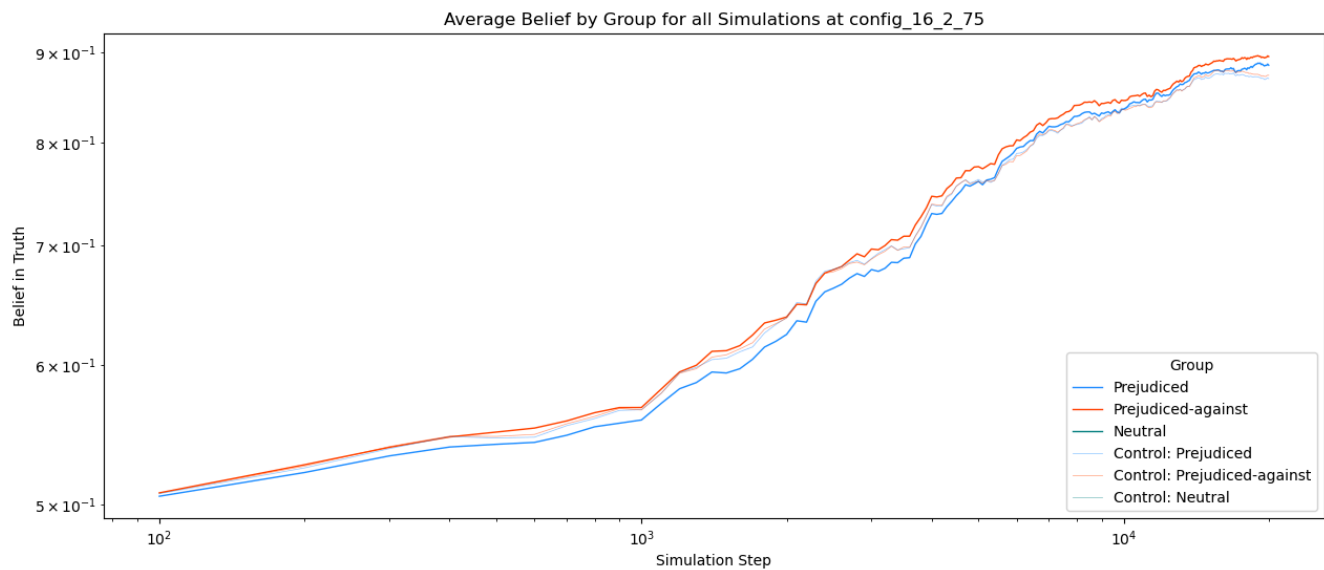


Figure 23. A line chart of average belief in truth by group for 100 simulations at size 16, 2 groups, 0.75 certainty in case of prejudice, alongside its control

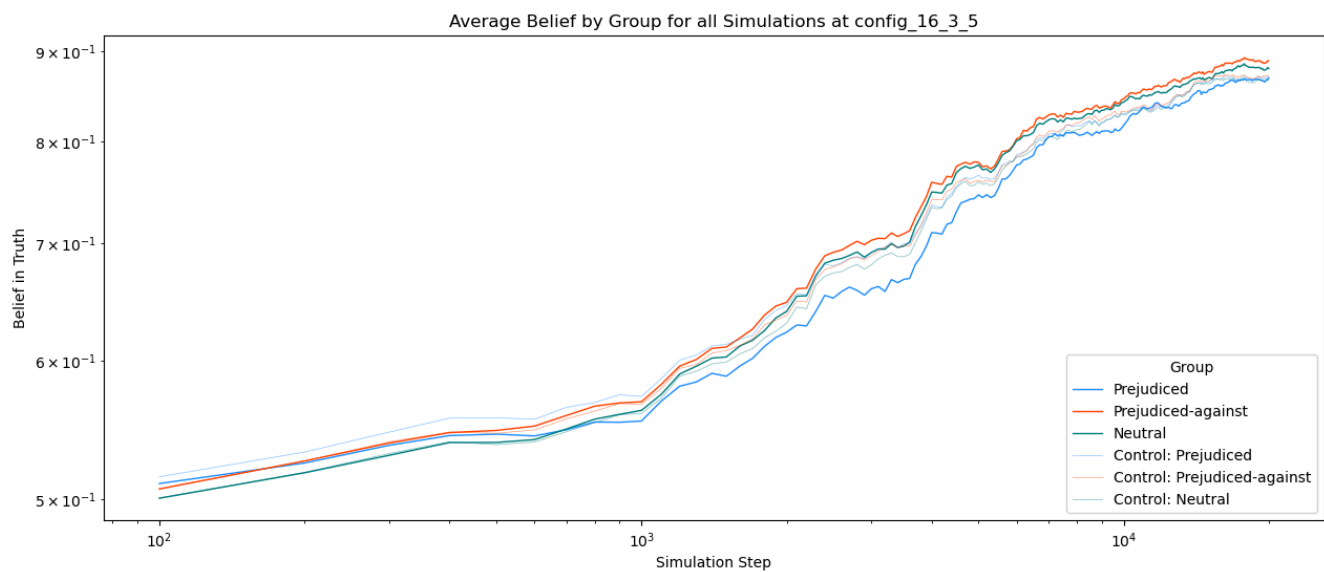


Figure 24. A line chart of average belief in truth by group for 100 simulations at size 16, 3 groups, 0.5 certainty in case of prejudice, alongside its control

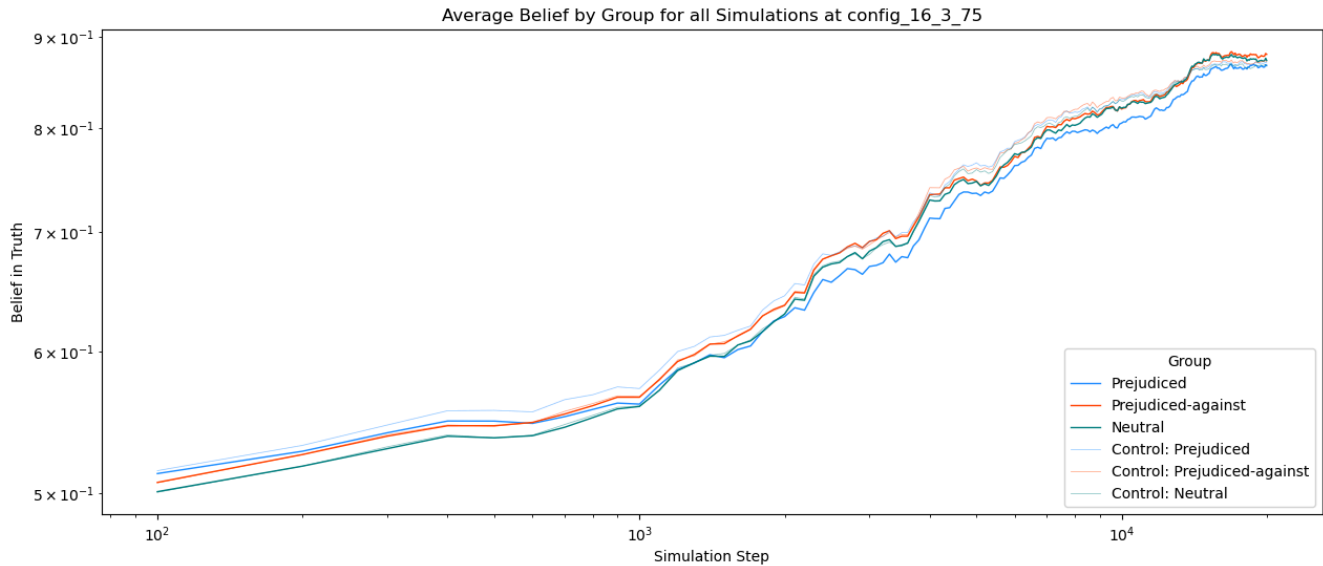


Figure 25. A line chart of average belief in truth by group for 100 simulations at size 16, 3 groups, 0.75 certainty in case of prejudice, alongside its control

Disregarding the control (for now), it appears that the nodes in the prejudiced group are still worse off in comparison to the nodes that they are prejudiced against, though the lines are close enough together in Figures 20 and 21, that the charts fails to effectively display the data at those simulation configurations.

Figures 24 and 25 visualize the previous, unexplained effect, where nodes in the neutral group are worse off than nodes in the prejudiced-against group. I suspect, however, that this is due to belief initialization, that perhaps I don't have enough simulations to reliably approximate the uniform distribution of beliefs at initialization. The same is not true for nodes in the prejudiced group—despite initializing with the greatest belief in truth, the nodes in this group consistently fall behind (epistemically) their peers.

4.4.1.1. A Startling Observation

Before I began testing to confirm or reject my hypotheses regarding the aforementioned group dynamics, I noticed something I could not, at the time, explain—the line charts displayed in Figures 18-25 indicate that nodes in the experiment simulations tend to have higher average belief at step 20,000 (the last step) than nodes in the control simulations. So, I wasn't just noticing that nodes in the prejudice-against group were typically better off than nodes in the prejudiced group, but that nodes in the prejudiced-against group often appeared better off than their counterparts in the control simulations! This was a new observation, so I, again, needed new data. I started re-running the last batch of simulations over a new set of 100 seeds. But first, I needed to perform the sign tests that I'd generated the current batch of data for.

4.4.2. Sign Test: Change in Average Belief by Group

Intervention: Transforming nodes in the prejudiced-against group into nodes in the prejudiced group (hypothetically)

Null Hypothesis: The average beliefs of nodes in the prejudiced group *are not* statistically significantly less than the average beliefs of nodes in the prejudiced-against group

Alternative Hypothesis: The average beliefs of nodes in the prejudiced group *are* statistically significantly less than the average beliefs of nodes in the prejudiced-against group

Test Results:

Sign Test Results					
	Config	k	n	Statistic	p-value
0	config_64_2_5	0	200	0.00	6.2e-61
1	config_64_2_75	3	200	0.01	8.3e-55
2	config_64_3_5	12	200	0.06	4.1e-42
3	config_64_3_75	154	200	0.77	1.0
4	config_16_2_5	0	200	0.00	6.2e-61
5	config_16_2_75	0	200	0.00	6.2e-61
6	config_16_3_5	1	200	0.01	1.3e-58
7	config_16_3_75	5	200	0.03	1.6e-51

Figure 26. Results of the sign test, indicating that the null hypothesis only holds for configuration, 'size: 64, groups: 3', CIP: 0.75.' The test addresses the question, "are nodes in the prejudiced group epistemically worse off than nodes in the prejudiced-against group?"

The sign test indicates that, indeed, at all configurations but one, the nodes in the prejudiced group are epistemically worse off than the nodes in the prejudiced-against group, to a statistically significant extent. And, the one configuration at which the null hypothesis holds, is that configuration where we understand prejudice to be the least: the size is 64, which means there is more good evidence (not proportionally, there are just more nodes to gather evidence), there are 3 groups, which means that only $\frac{1}{3}$ of the nodes in the network enact testimonial injustice, and CIP is set to 0.75, the lesser of the discounting-amounts. It is fair to conclude that, when there is enough testimonial injustice in a network, the nodes that do enact testimonial injustice are epistemically worse off than the nodes whose testimonies they discount, to a statistically significant extent.

4.5. Final Node-Belief Analysis

I, again, ran simulations at each of the eight configurations along with the 4 respective controls configurations, at 100 new seeds. I was looking to address the shocking observation I detailed in section 4.5.1.1, that networks with prejudice seemed to epistemically out-perform networks without prejudice. I didn't understand how this could be the case. To investigate the matter, I again plotted average belief over steps, this time disregarding groups to simplify the charts, which follow below.

4.5.1. Average Belief by Control/Experiment

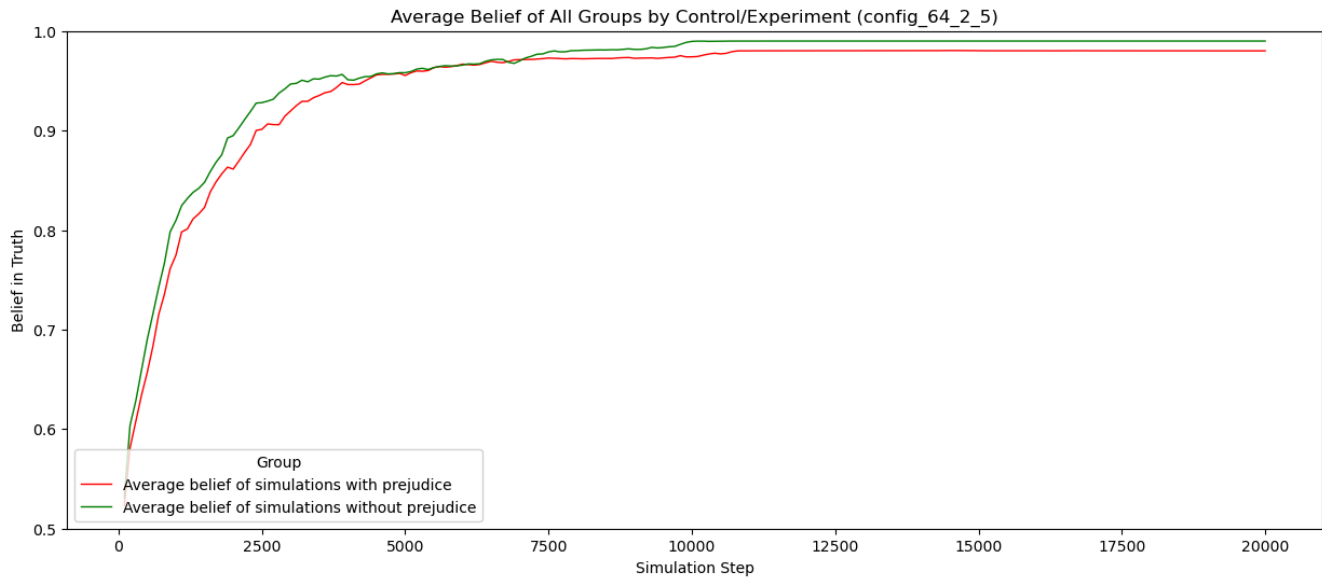


Figure 27. A line chart of average belief in truth by experiment / control for 100 simulations at size 64, 2 groups, 0.5 certainty in case of prejudice

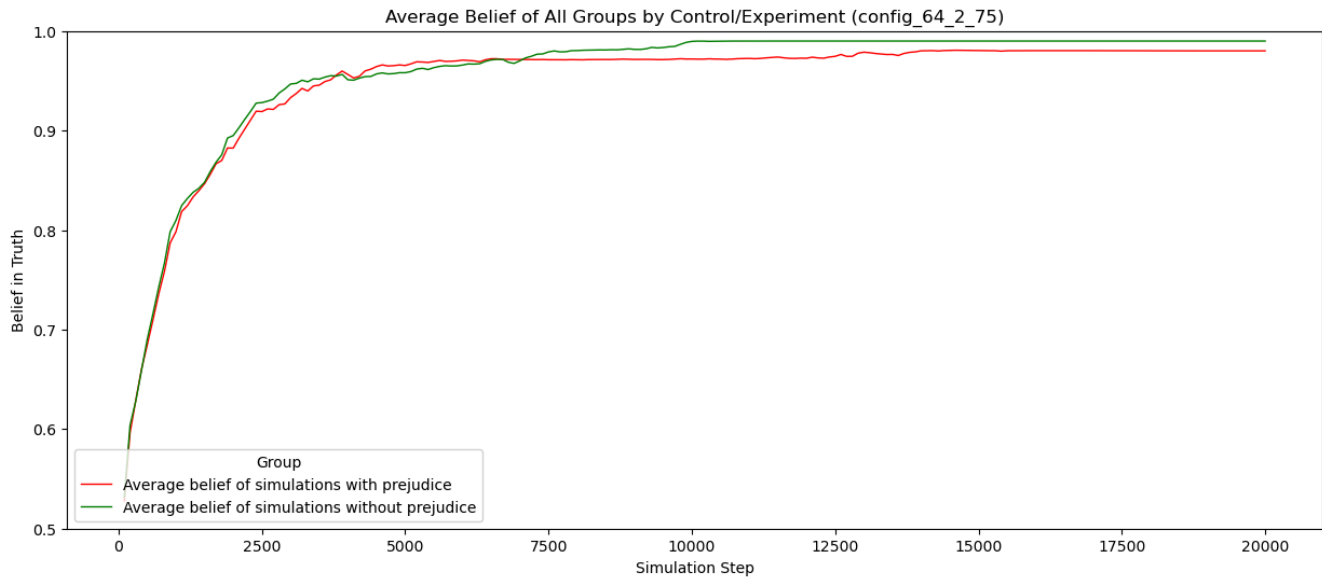


Figure 28. A line chart of average belief in truth by experiment / control for 100 simulations at size 64, 2 groups, 0.75 certainty in case of prejudice

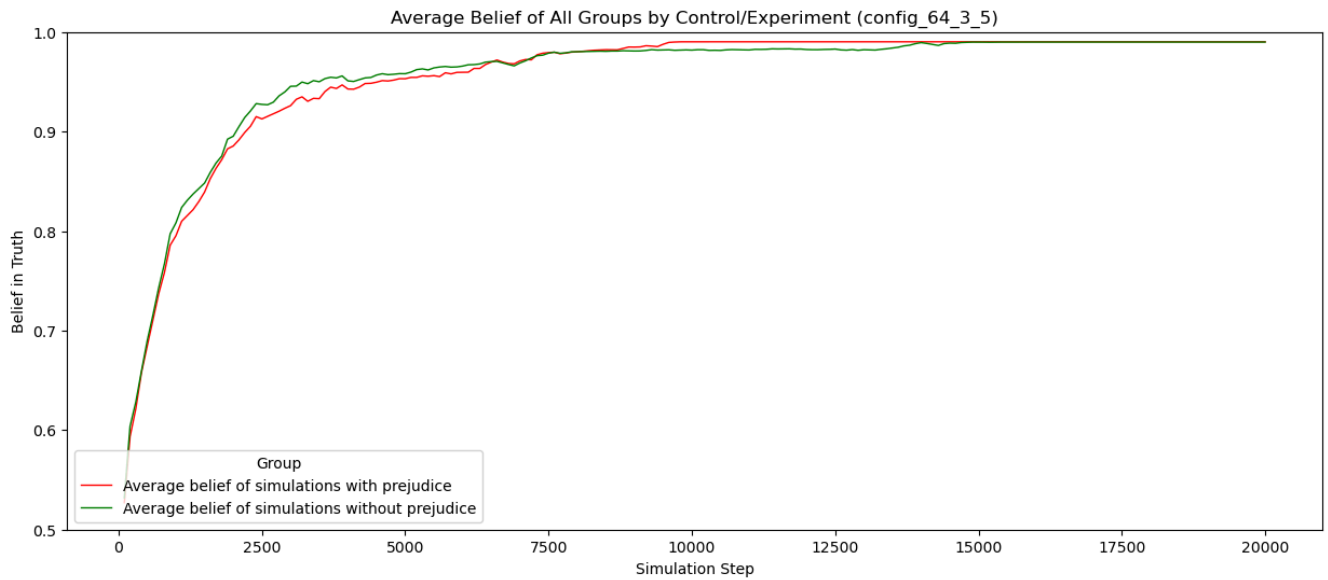


Figure 29. A line chart of average belief in truth by experiment / control for 100 simulations at size 64, 3 groups, 0.5 certainty in case of prejudice

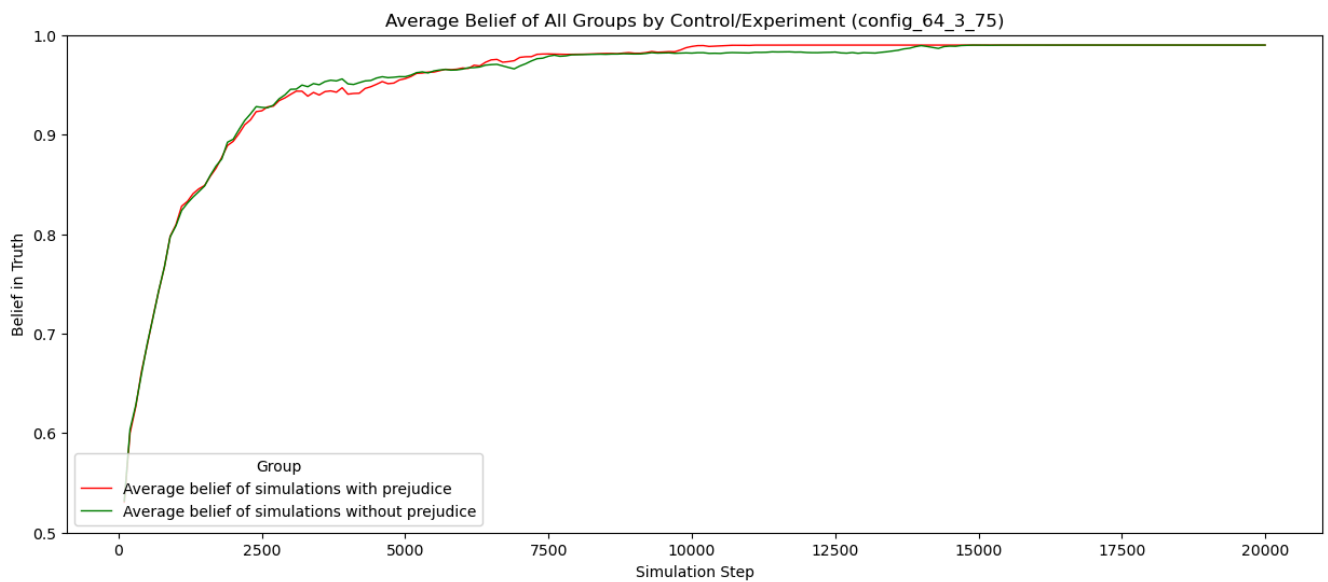


Figure 30. A line chart of average belief in truth by experiment / control for 100 simulations at size 64, 3 groups, 0.75 certainty in case of prejudice

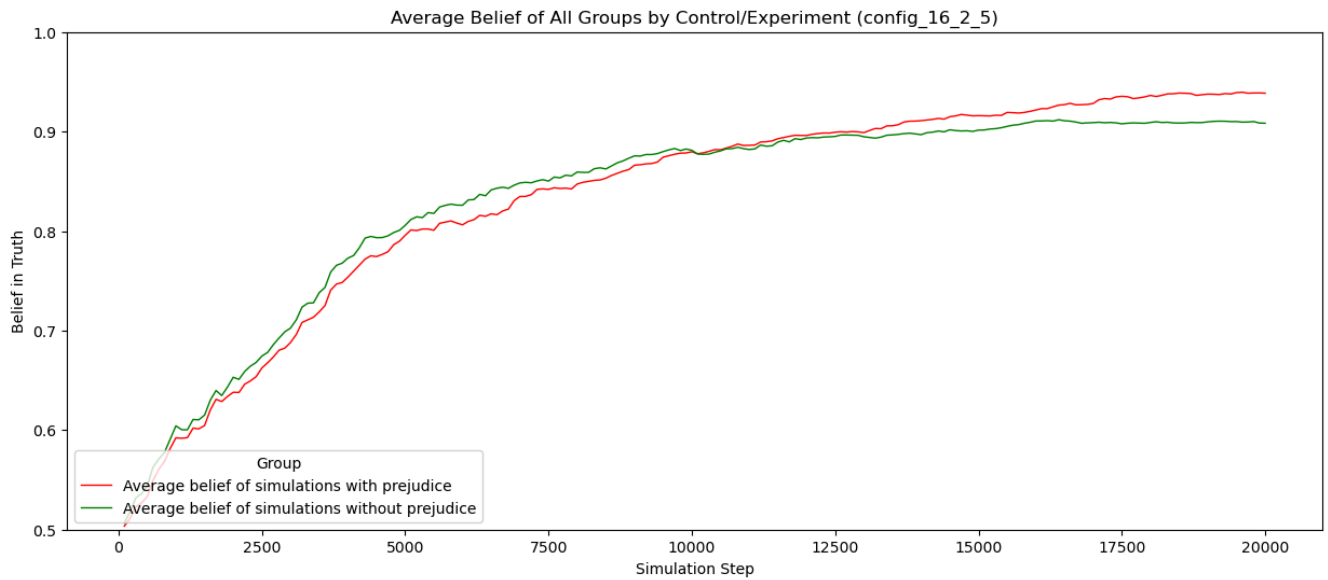


Figure 31. A line chart of average belief in truth by experiment / control for 100 simulations at size 16, 2 groups, 0.5 certainty in case of prejudice

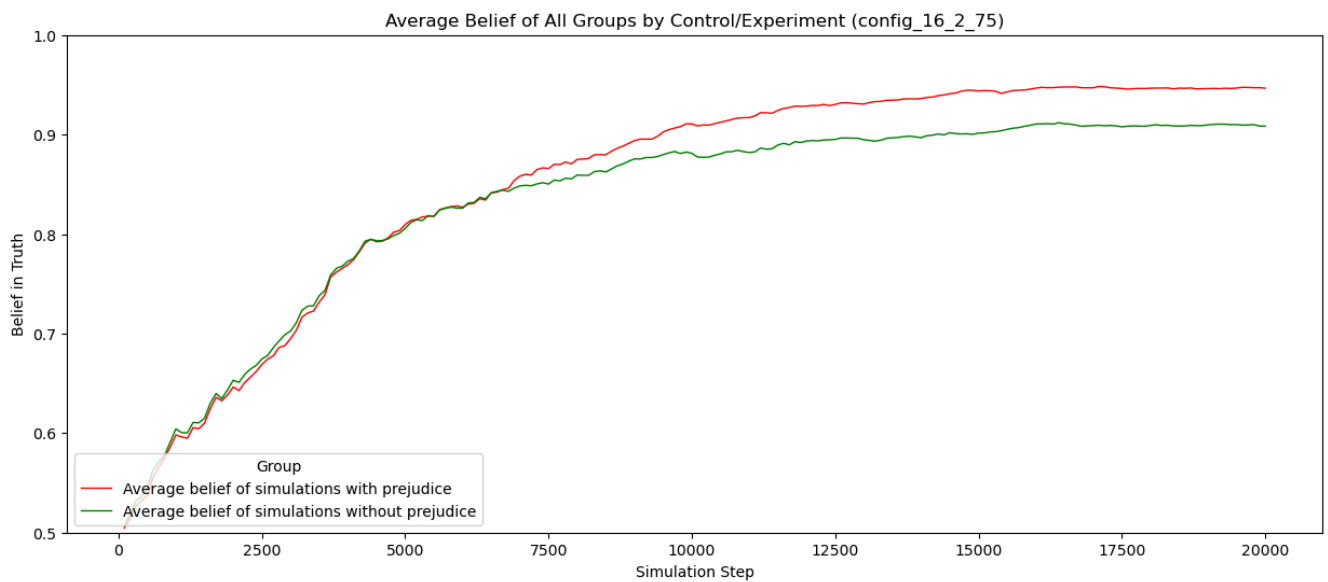


Figure 32. A line chart of average belief in truth by experiment / control for 100 simulations at size 16, 2 groups, 0.75 certainty in case of prejudice

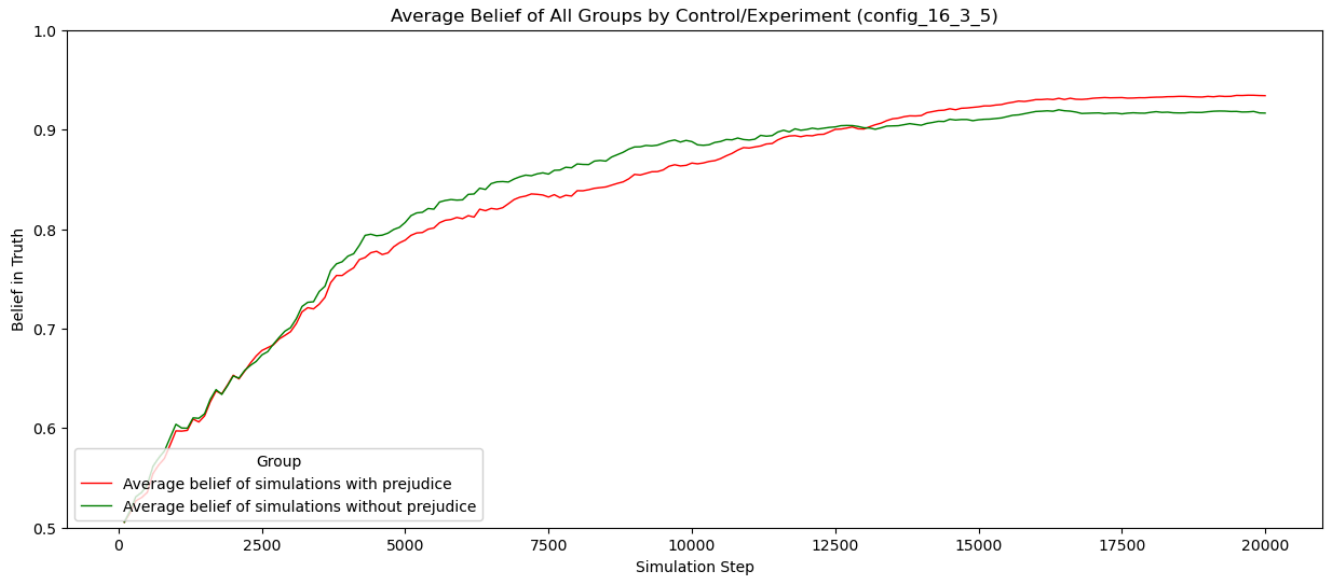


Figure 33. A line chart of average belief in truth by experiment / control for 100 simulations at size 16, 3 groups, 0.5 certainty in case of prejudice

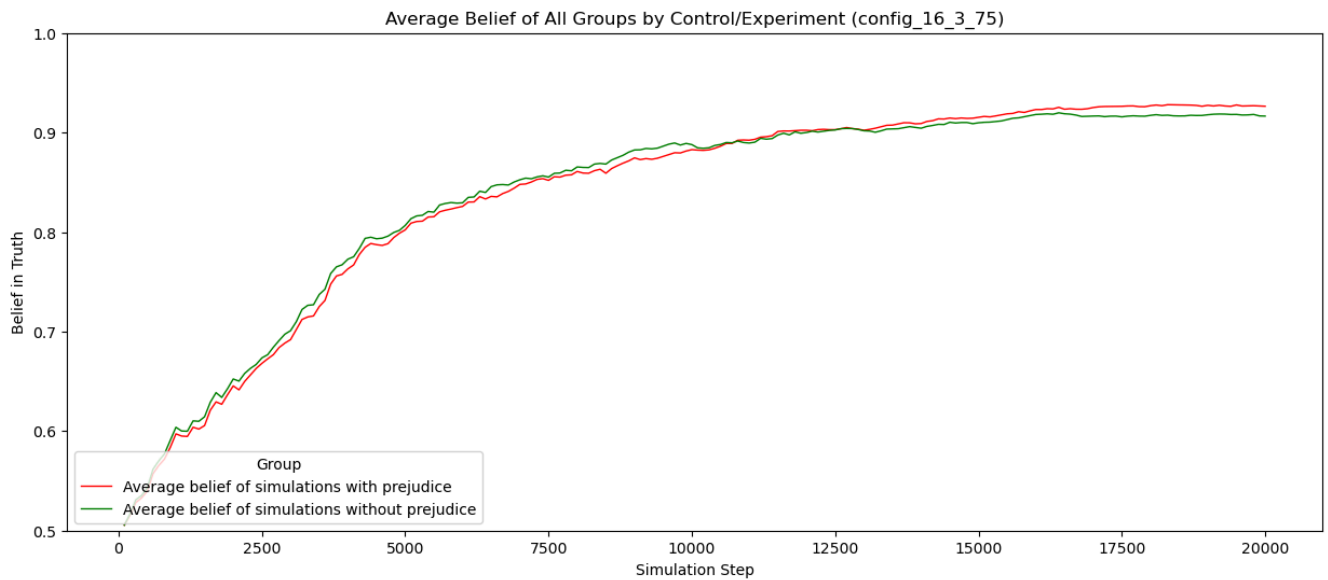


Figure 34. A line chart of average belief in truth by experiment / control for 100 simulations at size 16, 3 groups, 0.75 certainty in case of prejudice

The results are mostly as expected at size 64 (Figures 27-30), where the control networks seem to epistemically outperform the experiment networks, aside from a range from around step 10,000 to step 13,500, where the control network average dips below the experiment network average for those size 64 configurations with 3 groups (Figures 29-30).

The opposite is true at size 16 (Figures 31-34), where the experiment networks seem to outperform the control networks! In fact, at all configurations where size is 16, the experiment networks have greater average belief in truth than control networks at step 20,000.

4.5.2. Sign Tests: Change in Average Belief by Experiment/Control

To confirm what I was seeing, I ran two sign tests, asking “are nodes in networks with prejudice (networks that model testimonial injustice) epistemically worse off than nodes in networks without prejudice?” and vice-versa.

Intervention: Causing the nodes in the prejudiced group to enact testimonial injustice (turning on the prejudice)

Null Hypothesis: Nodes in networks *with* prejudice *are not* epistemically worse off than nodes in networks *without* prejudice

Alternative Hypothesis: Nodes in networks *with* prejudice *are* epistemically worse off than nodes in networks *without* prejudice

Test Results:

Sign Test Results					
	Config	k	n	Statistic	p-value
0	config_64_2_5	6	200	0.03	5.3e-50
1	config_64_2_75	35	200	0.17	1.1e-21
2	config_64_3_5	129	200	0.65	1.0
3	config_64_3_75	108	200	0.54	0.89
4	config_16_2_5	100	200	0.50	0.53
5	config_16_2_75	149	200	0.74	1.0
6	config_16_3_5	77	200	0.39	0.0007
7	config_16_3_75	92	200	0.46	0.14

Figure 35. A table of sign-test results, indicating that there is strong evidence to reject the null hypothesis at configurations 64_2_5, 64_2_75, and 16_3_5

The expected result occurs only at 3 of 8 simulation configurations— that being where nodes in networks with prejudice are epistemically worse off than nodes in networks without prejudice. And, at one of those three configurations (16_3_5, Figure 33), the experiment group is visibly better off than the control group at step 20,000, strongly suggesting that, if the simulations had been allowed to run longer, the null hypothesis would hold (at this configuration).

At this point, I was confused by what I was seeing. I ran another test, inverting the intervention.

Intervention: Preventing the nodes in the prejudiced group from enacting testimonial injustice (turning off the prejudice)

Null Hypothesis: Nodes in networks *without* prejudice *are not* epistemically worse off than nodes in networks *with* prejudice

Alternative Hypothesis: Nodes in networks *without* prejudice *are* epistemically worse off than nodes in networks *with* prejudice

Test Results:

Sign Test Results					
	Config	k	n	Statistic	p-value
0	config_64_2_5	194	200	0.97	1.0
1	config_64_2_75	165	200	0.82	1.0
2	config_64_3_5	71	200	0.35	2.5e-05
3	config_64_3_75	92	200	0.46	0.14
4	config_16_2_5	100	200	0.50	0.53
5	config_16_2_75	51	200	0.26	1.3e-12
6	config_16_3_5	123	200	0.61	1.0
7	config_16_3_75	108	200	0.54	0.89

Figure 36. A table of sign-test results, indicating that there is strong evidence to reject the null hypothesis at configurations 64_3_5 and 16_2_5

The unexpected outcome occurs at 2/8 configurations— at only 1 less configuration than the expected outcome. At this point, I felt very pressed to explain the results I was seeing, but could not figure it out.

To summarize: nodes in networks with prejudice are worse off, epistemically, on average, than nodes in networks without prejudice at 3/8 configs. (config_64_2_5, config_64_2_75, config_16_3_5). Networks without prejudice are worse than with prejudice 2/8 times. (config_64_3_5, config_16_2_75). And, at three of the configurations, there is no statistically significant difference (config_64_3_75, config_16_2_5, config_16_3_75).

I'm getting an effect at configs 64_3_5 and 16_2_75 that I cannot explain. However—looking at the chart for config 64_3_5 (Figure 29)—it doesn't really seem like the control group is worse off:

The control group seems significantly greater for most of the chart's path towards convergence to truth (most change in y-value). At around step 7,000 they seem to converge. Then, for an unknown reason, the experiment average overtakes the control average at around step 8,000. The experiment average stays above the control average until around step 14,000, where the control average eventually catches back up.

The sign test is saying that there are more snapshots of node-belief where the experiment group is greater than the control group, than where the experiment group is lesser. It does not speak to the magnitude of the difference.

When the control average is greater than the experiment average, the difference is larger than when the experiment average is greater than the control average. And, more growth along the y-axis occurs while the control average is greater— from 0.79 on the y-axis to 0.97. Then, while the control group sits at .98, the experiment overtakes it, increasing to 0.99. This is to say that, at configuration 64_3_5, the control group seems to have done a better job at getting *close* to the truth— a nuance lost on the

sign test.

Therefore, in order to explain the effect I was seeing, I focused on configuration 16_2_75, where the effect appeared most dramatic and striking.

4.5.3. An Investigation into Simulation Configuration 16-2-0.75

At this point, I speculated that using feed forward to solve the varying-simulation-lengths-problem might have been affecting my charts in an unforeseen manner: if it were the case that some simulations from the control group converged to falsehood, where those simulation's counterparts by seed did not, I would be averaging the whole control belief by a very low value for every step after its convergence, which would not be the case for the average experiment belief. In other words, network reliability could have a large impact on the chart under scrutiny (Figure 32).

4.5.3.1. Network Outcomes by Experiment/Control

In an attempt to confirm my speculation, as detailed in the previous section, I plotted counts of network outcomes by experiment/control. I was looking for a seed where that seed converged to truth under the experiment configuration, and to falsehood under the control configuration. Here is the chart:

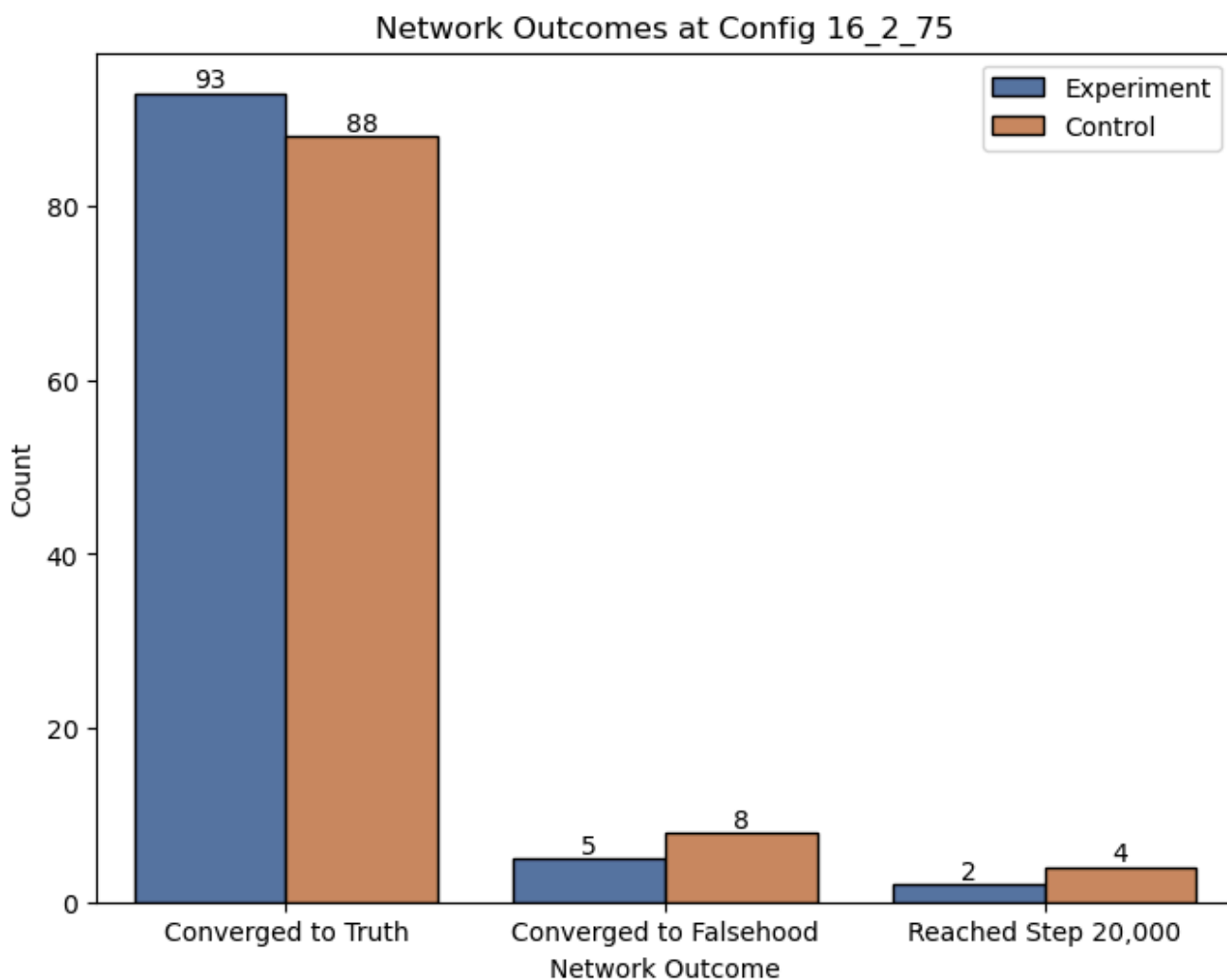


Figure 37. A bar chart of network outcome counts by experiment / control for simulation configuration: size 16, groups 2, CIP 0.75

As I'd suspected, the chart indicates that some (three) simulations converge to falsehood in the control set, that do not converge to falsehood in the experiment set. I still had no idea why this was occurring, but at least knew where to look next— at those three sims.

4.5.3.2. Node-Belief by Group, by Experiment/Control

Previously, in visualizing node-belief by group, I'd taken the beliefs of all the nodes by group, and calculated group-averages. Here, I visualize each individual node in a simulation as a line in a line chart. Each line chart visualizes the data from a single simulation— no averages are taken.

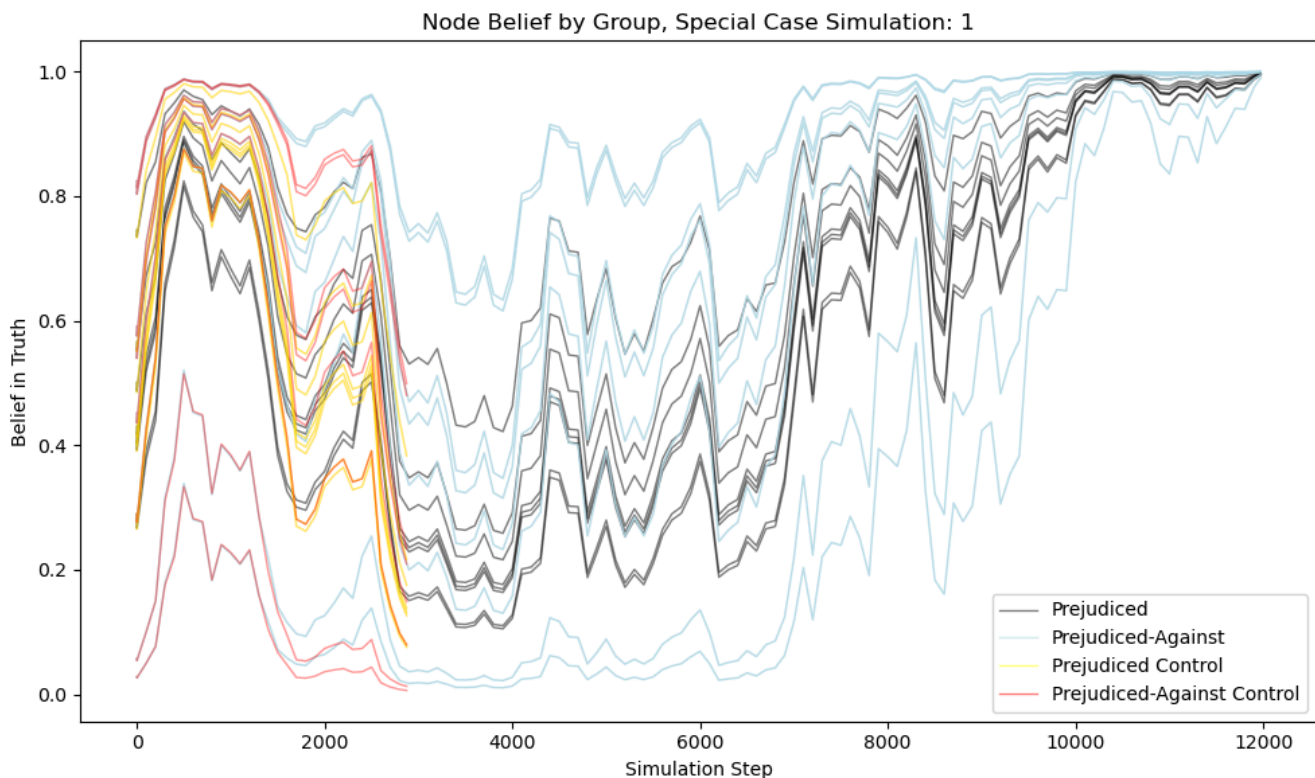


Figure 38. A line chart of each node's belief in truth, colored by group and experiment/control, at configuration: size 16, groups 2, CIP 0.75, seed 51117

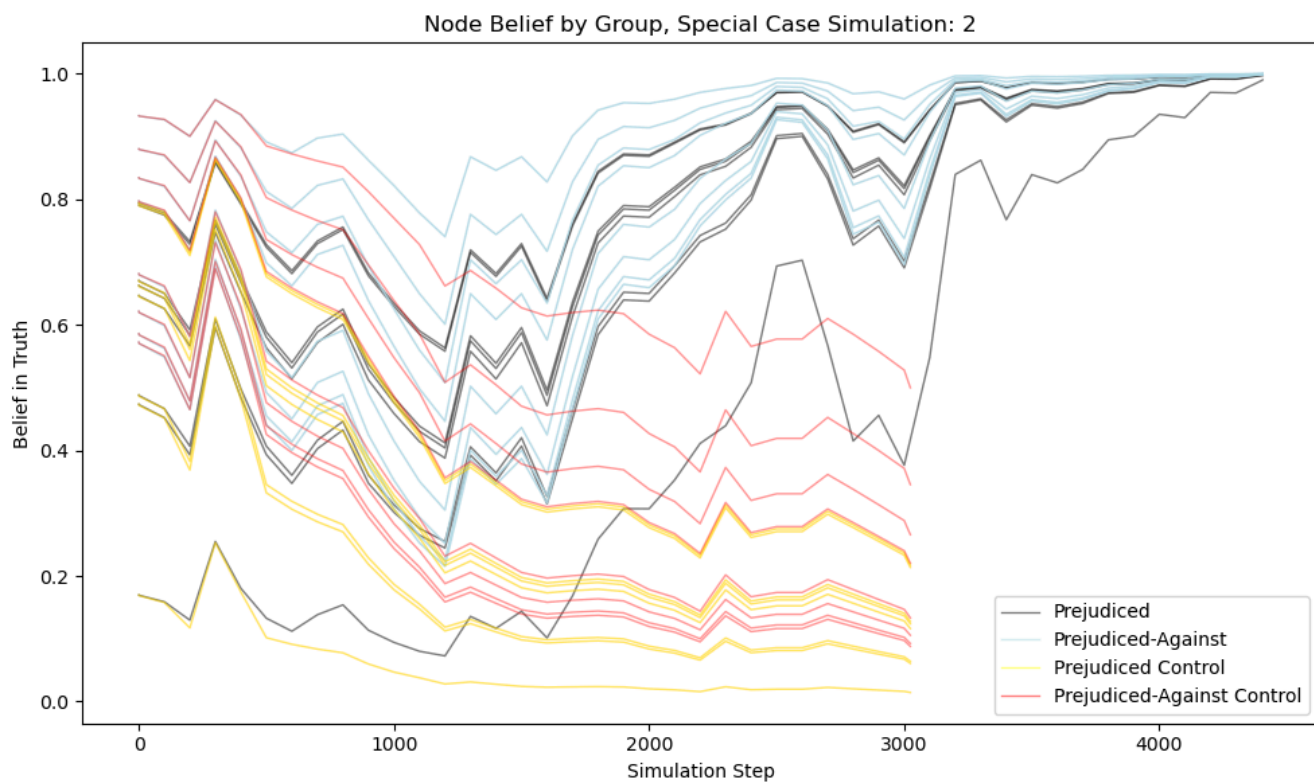


Figure 39. A line chart of each node's belief in truth, colored by group and experiment/control, at configuration: size 16, groups 2, CIP 0.75, seed 51192

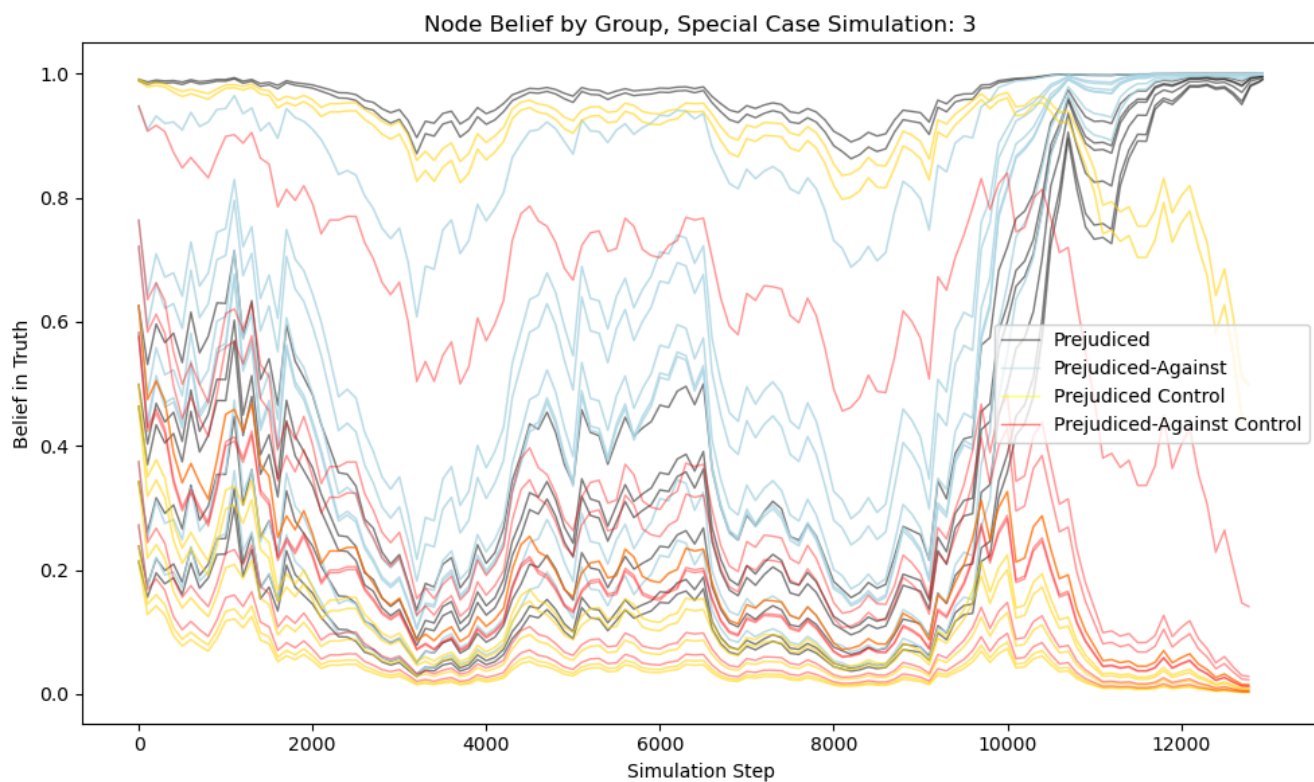


Figure 40. A line chart of each node's belief in truth, colored by group and experiment/control, at configuration: size 16,

I don't know why I didn't think of it when I was analyzing the figures in section 4.5.1, but, immediately, when I saw the beliefs of the individual nodes here, un-averaged, they screamed to me, "The Zollman Effect!" It became clear that the effect at hand wasn't a matter of prejudiced nodes doing better than non-prejudiced nodes (The Incidental-Filtering-by-Prejudice Effect)—it was a matter of overall network reliability. In the charts, we see that, whenever there is a down-spike in belief in truth (from misleading evidence), nodes in the control group are more greatly affected. This is especially apparent in Figure 39, where nodes in both the prejudiced and prejudiced-against groups show huge resiliency in contrast to the control from simulation step 1600, onwards.

To be clear, the Zollman effect describes the tradeoff (in epistemic networks) between network speed and network reliability by density, where more dense networks get to the truth faster, but more sparse (less dense) networks get there more reliably.¹¹ Looking back on the charts in section 4.6.1 (Figures 27-34), it makes sense that the effect appears so much more significant at size 16, as opposed to size 64—Rosenstock, O'Connor, and Bruner reproduce Zollman's simulations across more parameters, and report that, 'The Zollman effect is strongest for smaller networks, but as network size increases it drops off.'¹² And, although Rosenstock, Bruner, and O'Connor report that the parameter space under which the Zollman effect occurs is rather small,¹³ the parameters that I checked happened to be in that small space. This is particularly true of epsilon, which I only checked at 0.001—a small value that defines 'drug-B' as only marginally more effective than 'drug-A'. Because of this marginal difference, misleading evidence is more likely, and so provides the Zollman effect its required conditions.

Although I'd managed to attribute the startling observation described in section 4.5.1.1 to the Zollman effect, it was not immediately apparent to me how I'd *reproduced* the effect. After all, Zollman used network density as the independent (manipulated) variable in his experiment (comparing complete networks to less dense networks, such as ring networks), whereas I had merely implemented epistemic group-dynamics, or so I thought.

4.6. Testimonial Injustice as an Effector of Network Sparsity

As it turns out, I had conceptualized testimonial injustice in my simulations as, solely, a group dynamic because I had *intended* to model testimonial injustice as such. Instead, by using a network with no uncertainty as my control, I'd unknowingly modeled testimonial injustice as a group dynamic *and* as an effector of network sparsity; my independent variable came to include network density in relation to the control.

Take the following example of a network in my experiment:

¹¹ B. Ball, A. Koliousis, A. Mohanan, and M. Peacey, *Computational philosophy: reflections on the PolyGraphs project*.

¹² S. Rosenstock, C. O'Connor, and J. Bruner, *In epistemic networks, is less really more?*

¹³ *Ibid.*

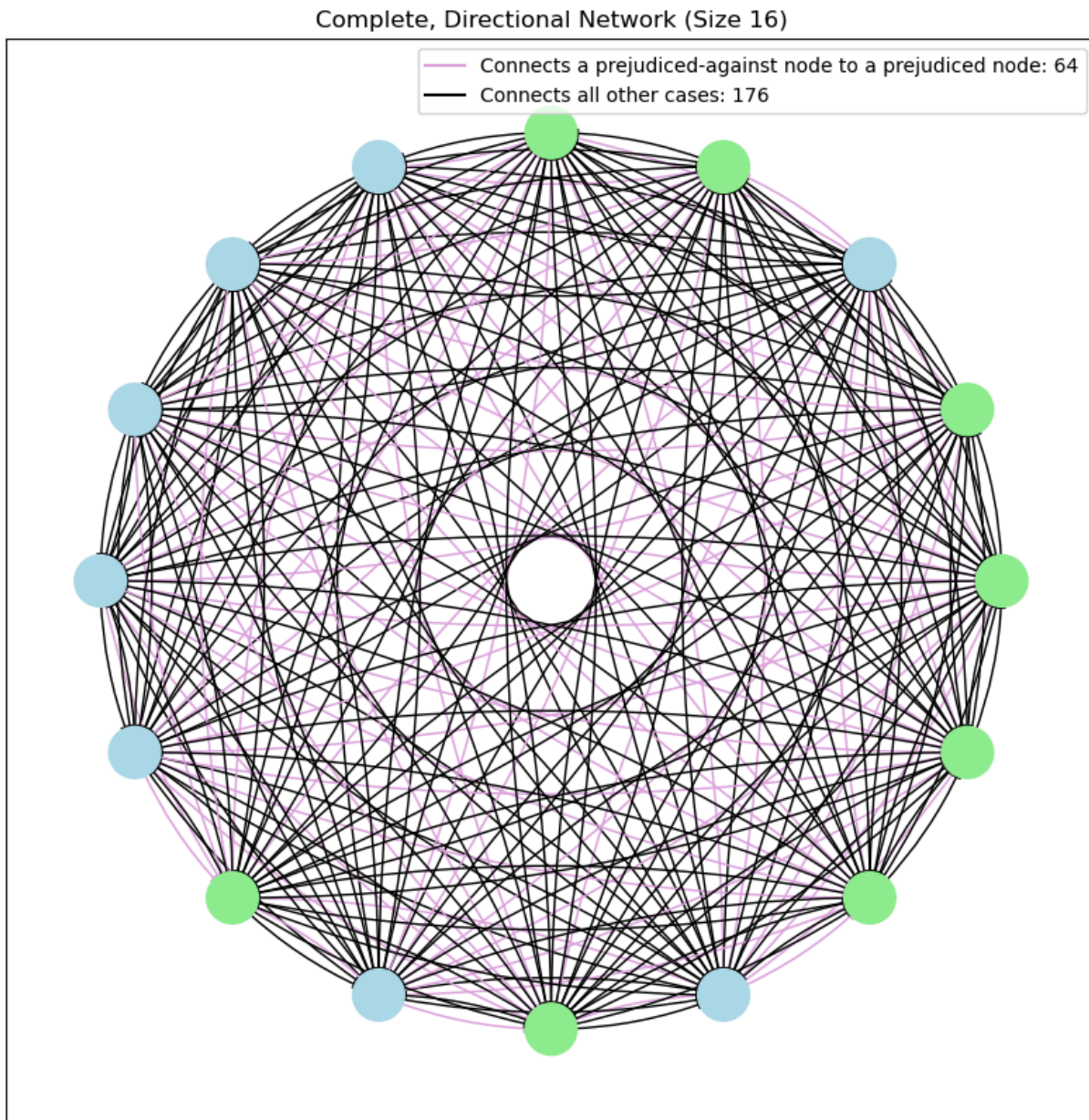


Figure 41. Visualization of a complete network at size 16, 2 groups, with edges colored where affected by testimonial injustice

Note here that, although the networks in my experiment are not explicitly directional (one-way connections between nodes), they are effectively directional. This is because instances of testimonial injustice depend on a sender-receiver relationship. Therefore, I have visualized the example network as such. This was necessary to color only those connections that are affected by testimonial injustice, and it is in considering these colored connections that I manage to explain my reproduction of the Zollman effect.

It seems right to conceptualize the edges in my network as representations of the transmission of evidence, but this is slightly reductive. Suppose that, in all cases, certainty was set to 0, and all evidence was consequently completely discarded. Topographically, the network would still have its edges, but effectively—behaviorally—it would have none. The edges, then, represent the

transmission *and acceptance* (updating-on) of evidence.

Therefore, the plum edges in Figure 41 are effectively *partial* edges. If CIP was set to 0, it would be like they weren't there. Instead, CIP is set somewhere between 1 and 0, and so the plum edges are somewhere between 'being there fully' and 'not being there at all.'

Given enough of these partial edges, we approximate the behavior of a less-dense network (as opposed to if all edges were 'fully-there'). Therefore, as testimonial injustice (as I've modeled it) increases, network density decreases. This is how I inadvertently reproduced the Zollman effect.

5. Conclusion

5.1. Evaluation & Further Work

Although I did not explicitly intend for my independent variable to involve network density, I do not believe that the experiment is invalidated as a result. It can help to think about the experiment and control groups flipped here, as, in a way, I've tested for half of the question, 'what happens to a network, epistemically, when we *remove* testimonial injustice from that network?'

This is because there are two ways to remove testimonial injustice from a network, or more precisely, to change a node so that it no longer enacts testimonial injustice— one for each of the necessary conditions on testimonial injustice:

To enact testimonial injustice, you must 1. have uncertainty about some testimony, and 2. have that uncertainty on account of the testifier's membership to some group. So, in my experiment (thinking about it as its inverse), I stopped nodes from enacting testimonial injustice by removing their uncertainty, altogether.

Instead of simply preventing nodes from enacting testimonial injustice by removing their uncertainty, I could have done so by ensuring that nodes did not distribute their uncertainty by *group*, thereby failing the second aforementioned necessary condition. In other words, my control (back to non-inverse) shouldn't have only covered 'no testimonial injustice, as in no uncertainty,' but also, 'no testimonial injustice, as in evenly distributed uncertainty.' This alternative would have better isolated my independent variable by controlling for network density.

Therefore, the next clear step in this research is to re-run the experiment using this alternative control condition. The custom operation I've written needs only be slightly modified to run the proposed control simulations. Additionally, expanding the parameter space under which the effects of testimonial injustice are investigated would further enhance the robustness of the findings, and mitigate the Zollman effect's impact on the results.

5.2. On Testimonial Injustice

Although the results of my experiment demonstrate that, given certain conditions, testimonial injustice can increase the reliability of an epistemic network, the results do not suggest that we ought to enact testimonial injustice. In section 4.4.2, we saw that, taken as individual agents, the nodes that enact testimonial injustice are significantly worse-off (epistemically) than the nodes that do not, at almost all configurations checked. In considering one's own epistemic interest, testimonial injustice is clearly bad.

This claim is less obviously true in considering the epistemic interest of the whole network, given the Zollman effect, but the instances of increased network reliability do not warrant testimonial injustice:

1. Rosenstock, O'Connor, and Bruner suggest that, if we are to take advice from the simulation results of epistemic network models, we must be sure that the results in question are robust across a wide range of network parameters and topologies.¹⁴ They propose this condition due to the reductiveness / idealization of epistemic network models— we can not say for certain when models of epistemic networks perfectly correspond with actual epistemic networks, so unless some modeled effect occurs across a wide range of networks, we cannot assume that the effect will occur in a given actual epistemic network. Therefore, because the Zollman effect is not very robust—particularly in respect to network-size and epsilon—we have good reason to doubt that the effect occurs in actual epistemic networks.
2. Even if we suppose that we are certain that the actual-network in question falls under the parameter space where the Zollman effect occurs in modeled-networks, we need not enact testimonial injustice to incur the network-wide benefits of reduced density. As proposed for a new control group in the previous section, we might instead distribute our uncertainty uniformly across our neighbors.

While testimonial injustice may enhance network reliability under certain conditions, the epistemic cost to prejudiced nodes, decreased overall network speed, and lack of robustness in the Zollman effect collectively suggest that testimonial injustice typically has a negative impact on group knowledge, and often significantly so, as indicated by the tests performed in section 4. I conclude that testimonial injustice remains epistemically unwarranted.

¹⁴ *Ibid.*

References

- A. Barabási, *Network science*. Cambridge: Cambridge University Press, 2016.
<http://www.networksciencebook.com/>
- A. Tilevik, "How to choose an appropriate statistical test." YouTube, TileStats, 2021.
<https://www.youtube.com/watch?v=dYJLUvo0Q6g&list=LL&index=16>.
- B. Ball, A. Koliouisis, A. Mohanan, and M. Peacey, "Computational philosophy: reflections on the PolyGraphs project," *Humanities and Social Sciences Communications*, vol. 11, no. 1, Jan. 2024, doi: <https://doi.org/10.1057/s41599-024-02619-z>.
- C. O'Connor and James Owen Weatherall, *The Misinformation Age: How False Beliefs Spread*. Yale University Press, ch. 2, 2019.
- D. Montgomery and Runger, *Applied statistics and probability for engineers*. Hoboken, NJ: Wiley, p.227
- K. Zollman, *The communication structure of epistemic communities*. *Philosophy of Science*, 74(5), pp.570–590, 2007.
- M. Fricker, "Epistemic Injustice," *Epistemic Injustice: Power and the Ethics of Knowing*, Jun. 2007, doi: <https://doi.org/10.1093/acprof:oso/9780198237907.001.0001>.
- S. Rosenstock, C. O'Connor, and J. Bruner, *In epistemic networks, is less really more?* *Philosophy of Science*, 84(2), pp.234–252, 2017.

